



Luncheon talk:

Beattie, Federrath, Klessen, Cielo, Bhattacharjee, 2025 (*Nature Astronomy*). The spectrum of magnetized turbulence in the interstellar medium

Supernova remnants as engines of non-Kolmogorov galactic turbulence (and why we need to throw away old ideas applied to ISM turbulence)

ITC Colloquium, Harvard University

James Beattie

Fellow / Research Associate

Canadian Institute for Theoretical Astrophysics / Princeton

In collaboration: Isabelle Connor (UCSC), Anne Noer Kolborg (UCSC), Enrico Ramirez Ruiz (UCSC), Amitava Bhattacharjee (Princeton), Christoph Federrath (ANU), Ralf Klessen (UH), Salvatore Cielo (LRZ)

Beattie + (2025; ApJ) So long Kolmogorov: the forward and backwards cascades in a supernovae-driven, multiphase ISM
Connor, ***Beattie*** + (2025, ApJ) Cascading from the winds to the disk: universality of supernovae-driven turbulence in different galactic ISMs
Beattie (2026; submitted ApJL) Supernovae drive large-scale incompressible turbulence from small-scale instabilities

Roadmap for this colloquium

1. Discuss vanilla K41 turbulence in a pedagogical manner & turbulence in our galaxy.
2. Supernova-driven turbulence — how to fit them into the cascade? Simulations.
3. Spectra, cascade directions.
4. Where does the incompressible turbulence come from — shell instabilities and spectra model.

The ISM: A laboratory for interesting nonlinear physics

- Multiphase trans-to-supersonic compressible plasma
- $Re = UL/\nu \gg 1 \implies$ highly-turbulent
- Turbulence driven by a zoo of sources
 - Gravitational instabilities, SN explosions, stellar feedback...
- Pervaded by ultra-relativistic particles with non-negligible energy densities (peaked number density in GeV – TeV range)
- Non-Gaussian gas-density structures related to star formation (multi-scale filaments, ribbons, striations, sheets)
- Dynamically important, maintained magnetic fields
- $Rm = UL/\eta \gg 1 \implies$ highly-turbulent magnetic fields
-

Krumholz & McKee (2005); Federrath & Klessen (2012); Tritsis+(2016); Xu & Lazarian (2016); Soler & Hennebelle (2017); Squire & Hopkins (2017); Mocz & Burkhardt (2018), Burkhardt & Mocz (2018); Heyer+(2012); Tritsis+(2018); Hu+(2019); Heyer+(2020); Krumholz+(2020); Beattie & Federrath (2020); Beattie+(2020); Körtgen & Soler (2020); Skolidis & Tassis (2020); Federrath+(2021); Burkhardt (2021); Barreto-Mota+(2021); Hopkins+(2022); Beattie+(2022); Fielding+(2022); Sampson+ (2022); Kriel+(2022); Galishnikova+(2022); Beattie+(2023)

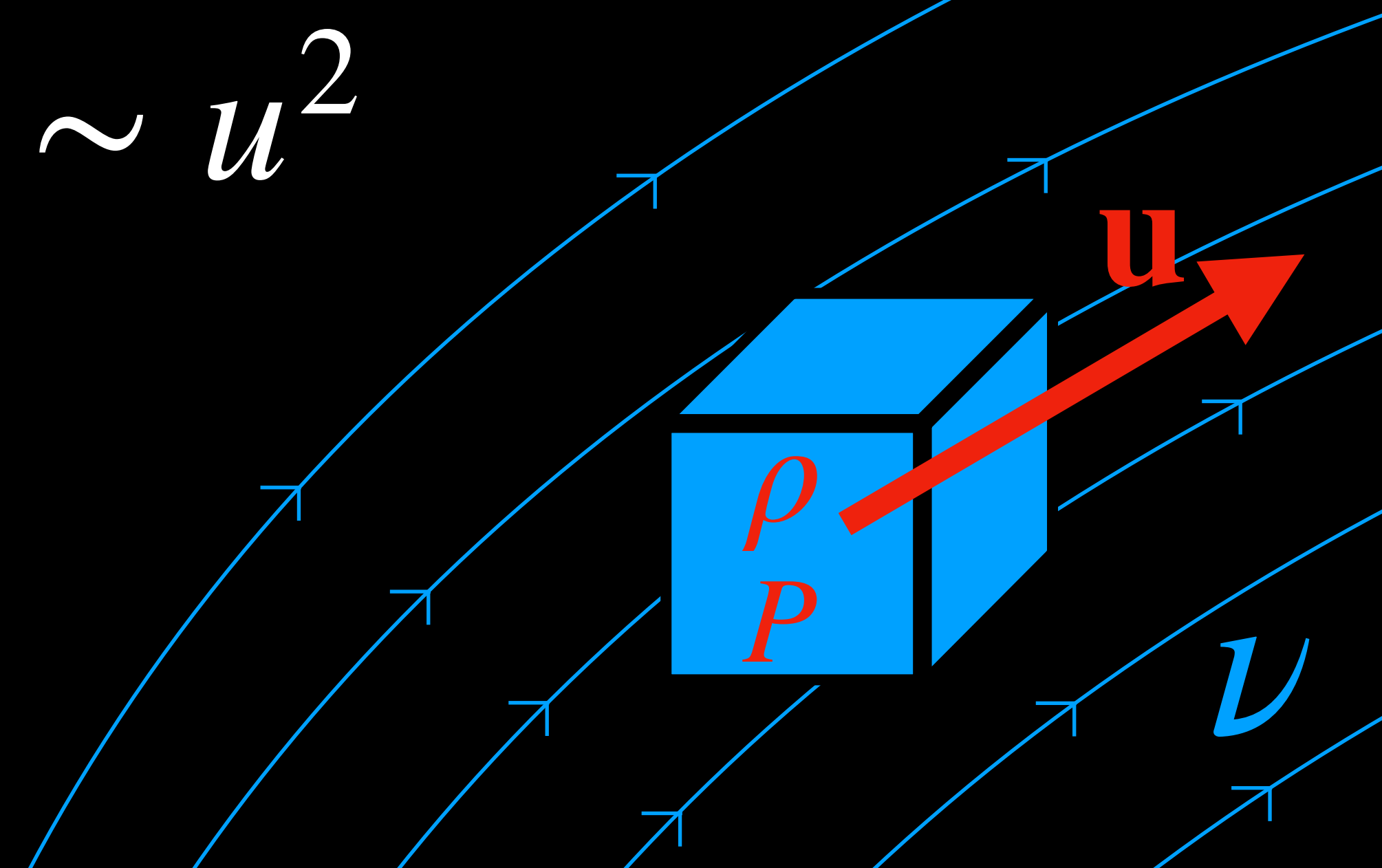
Turbulence

What is it (hand wavy)?

Momentum conservation for a hydrodynamical fluid element
(first moment of Boltzmann eq. for monoatomic plasma slightly out of LTE)

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P\mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$
$$\mathbf{u} \otimes \mathbf{u} = u_i u_j \sim u^2$$

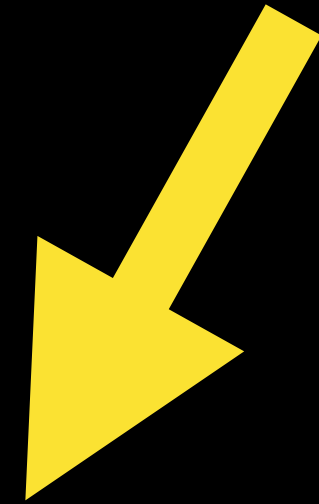
$$\lambda_{\text{mfp}}/L \ll 1$$



Turbulence

What is it (hand wavy)?

Reynolds stress



$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P\mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$
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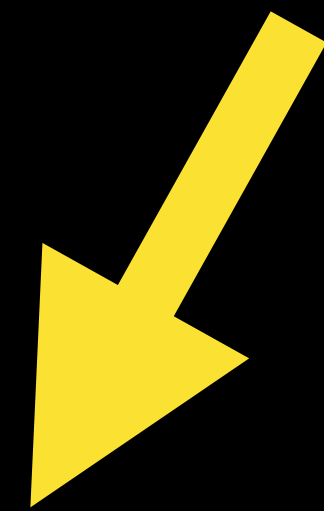


Viscous stress

Turbulence

What is it (hand wavey)?

creating nonlinearity



$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla \cdot P \mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$
$$\mathbf{u} \otimes \mathbf{u} = u_i u_j \sim u^2$$



Smooths out nonlinear things in the fluid

Turbulence

What is it (hand wavy)?

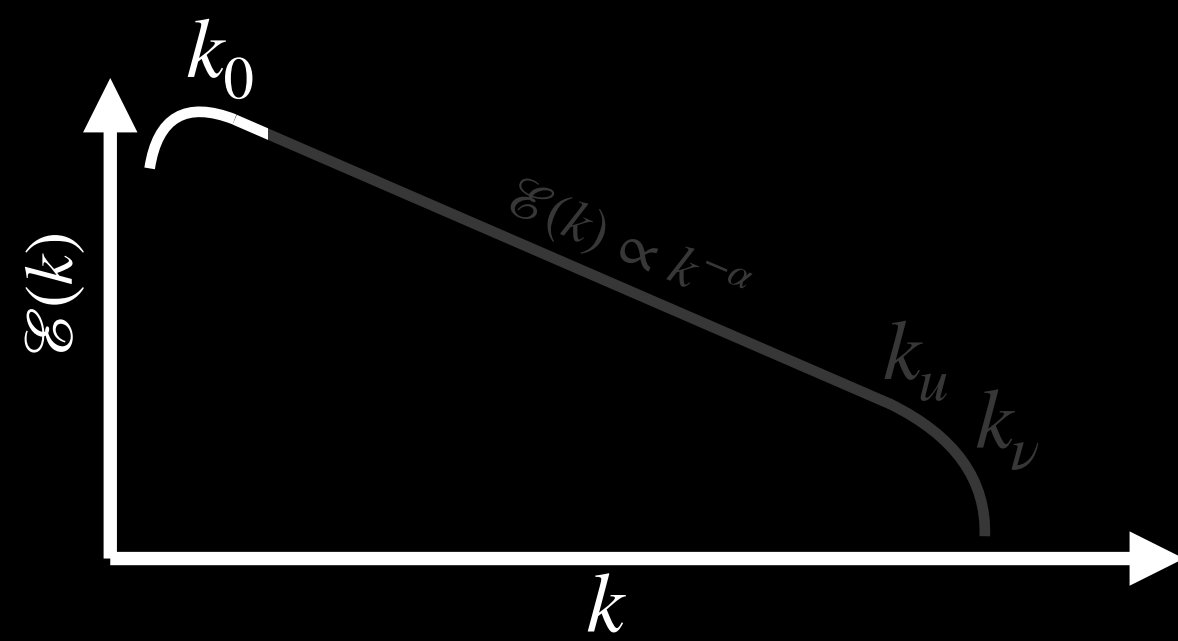
$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = - \nabla \cdot P \mathbb{I} + 2\nu \nabla \cdot (\rho \mathcal{S})$$

Creating nonlinear things in the fluid

$$\text{Re} = \frac{|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|}{|2\nu \nabla \cdot (\rho \mathcal{S})|} \sim \frac{UL}{\nu}$$

Smoothing out nonlinear things in the fluid

Turbulence Cascade



Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

inspired by Shukurov, 2011

$$\begin{aligned} \text{WIM: } \text{Re} &\sim 10^7 & \lambda_{\text{mfp}} &\sim 5 R_{\oplus} \\ \text{WNM: } \text{Re} &\sim 10^7 & \lambda_{\text{mfp}} &\sim 1.3 \text{ AU} \\ \text{CNM: } \text{Re} &\sim 10^{10} & \lambda_{\text{mfp}} &\sim 0.3 \text{ pc} \end{aligned}$$

Ferrière, 2020;

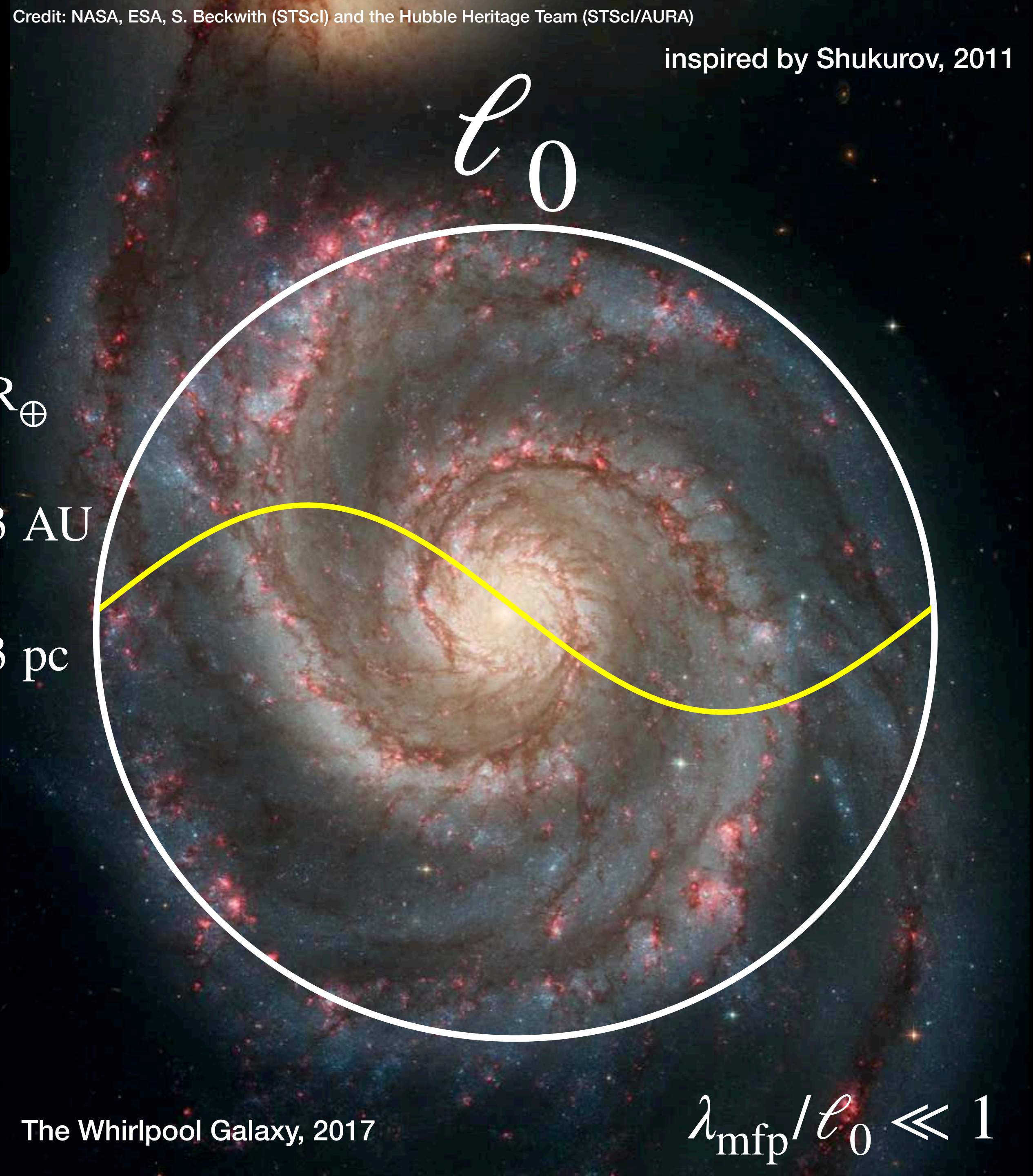
The quadratic nonlinear term

$$|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|$$

dominates on large scales, ℓ

The Whirlpool Galaxy, 2017

$$\lambda_{\text{mfp}} / \ell_0 \ll 1$$



Turbulence

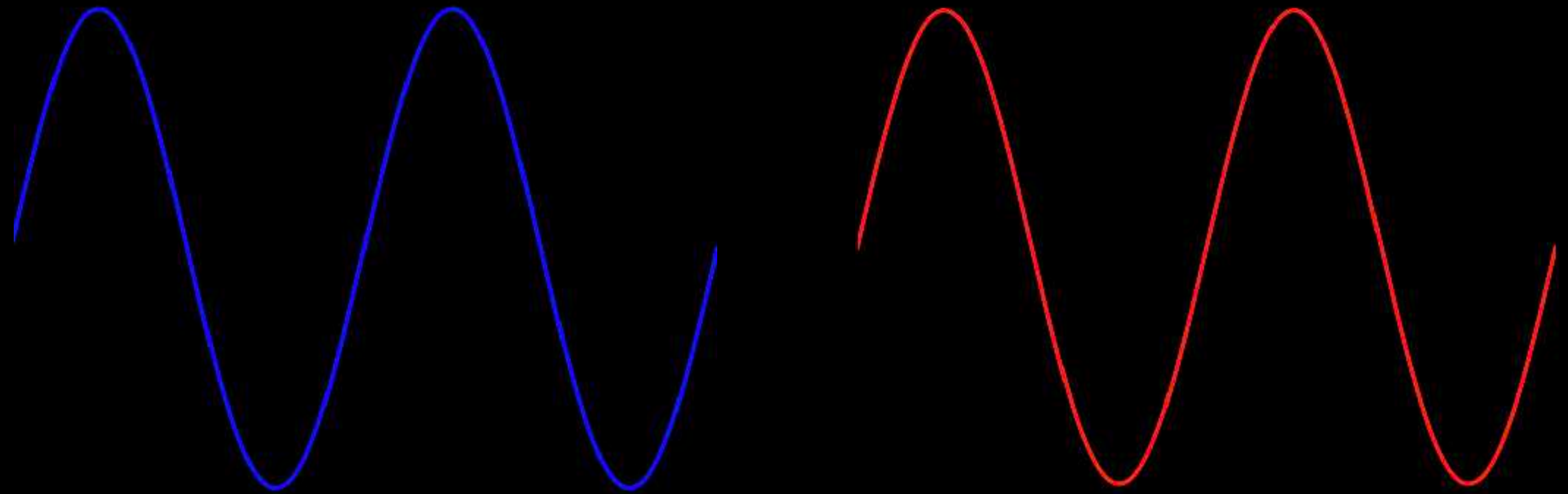
What does $\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$ want to do?

$$k_1 = k_2$$

Consider two waves

$$u_1(x) = \sin(k_1 x)$$

$$u_2(x) = \sin(k_2 x)$$



The nonlinear term creates a new wave

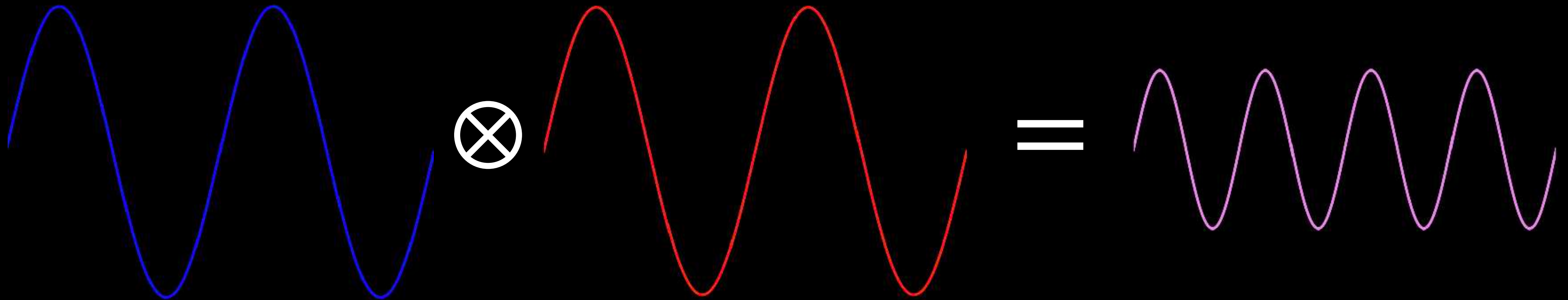
$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) \sim u_1 \partial_x u_2 = k_2 \sin(k_1 x) \cos(k_2 x) \propto \sin(k_3 x)$$

$$k_1 + k_2 = k_3 \quad (\text{momentum conservation})$$

Turbulence

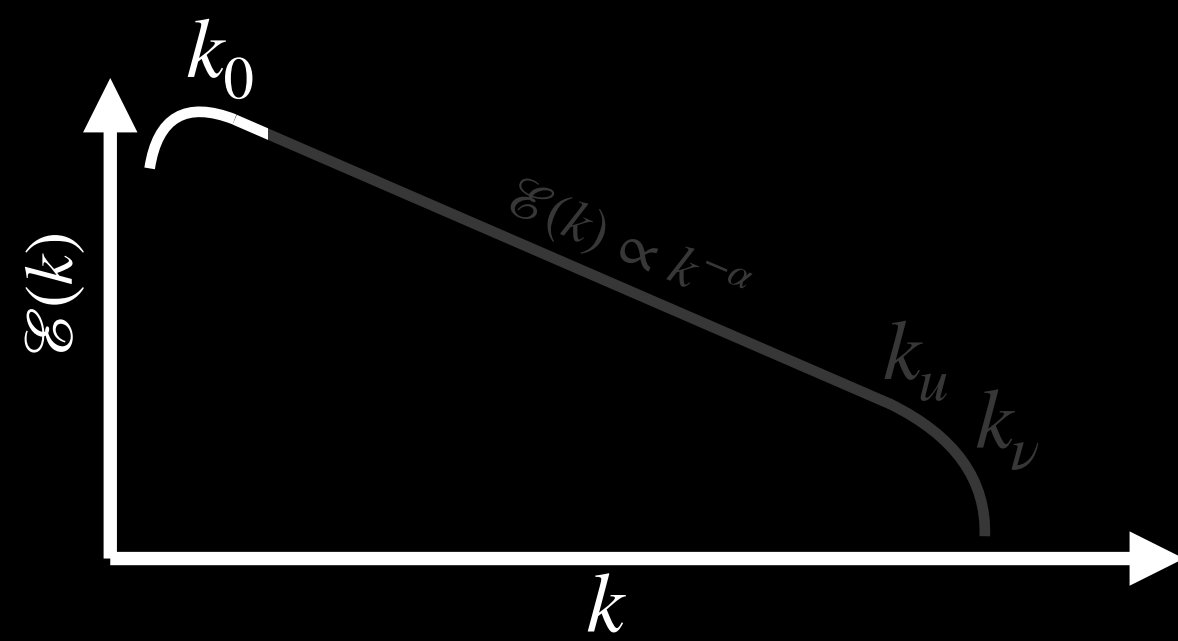
What does $\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$ want to do — cascade!!!

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})$$



$$k_1 + k_2 = k_3$$

Turbulence Cascade



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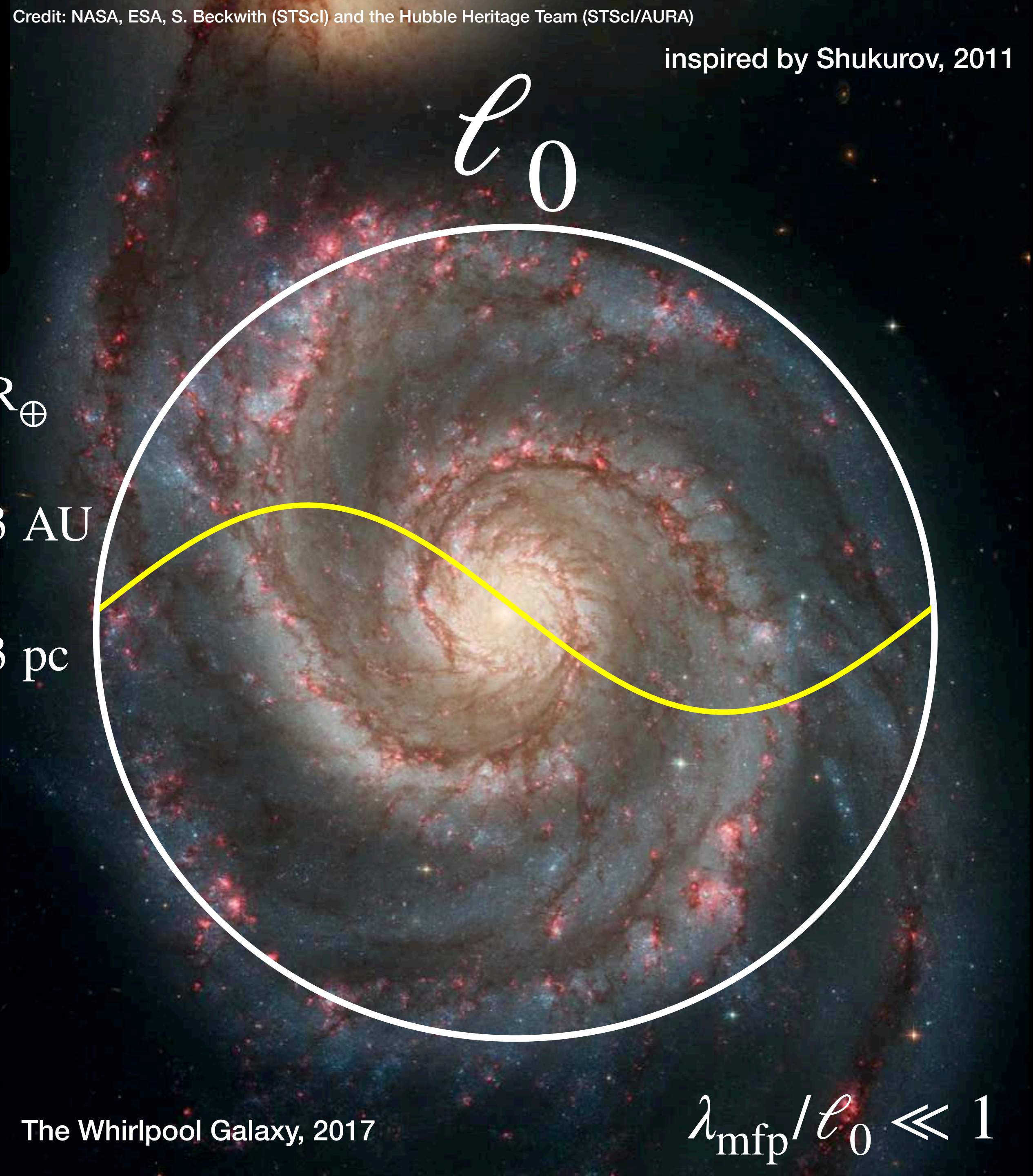
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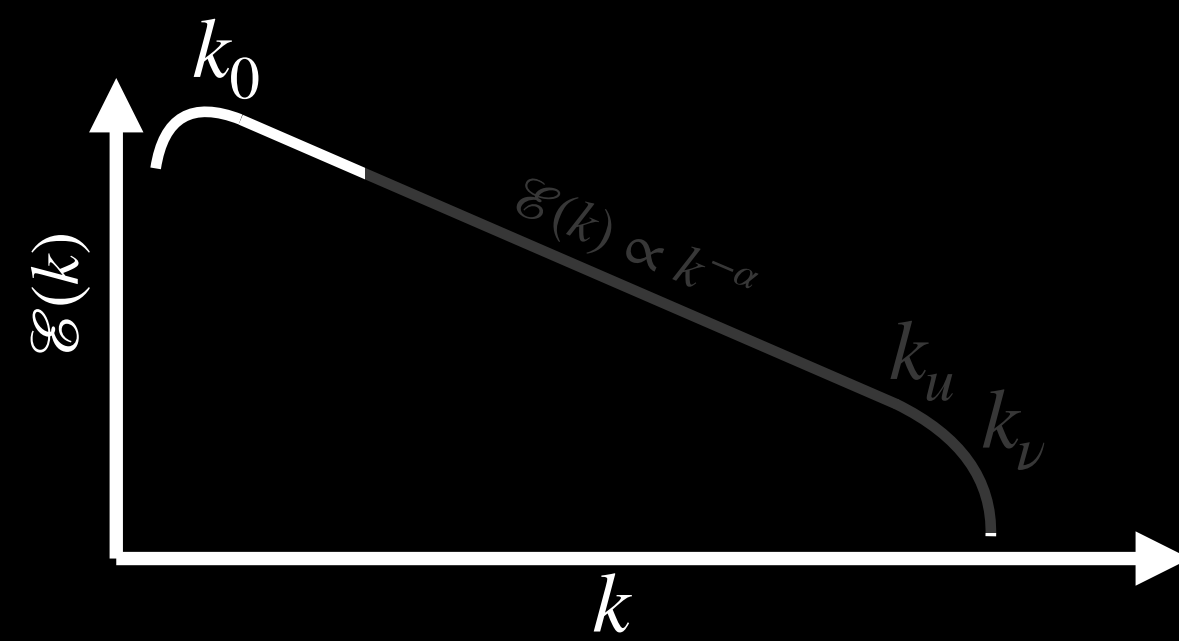
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The Whirlpool Galaxy, 2017

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Turbulence Cascade



The quadratic nonlinear term

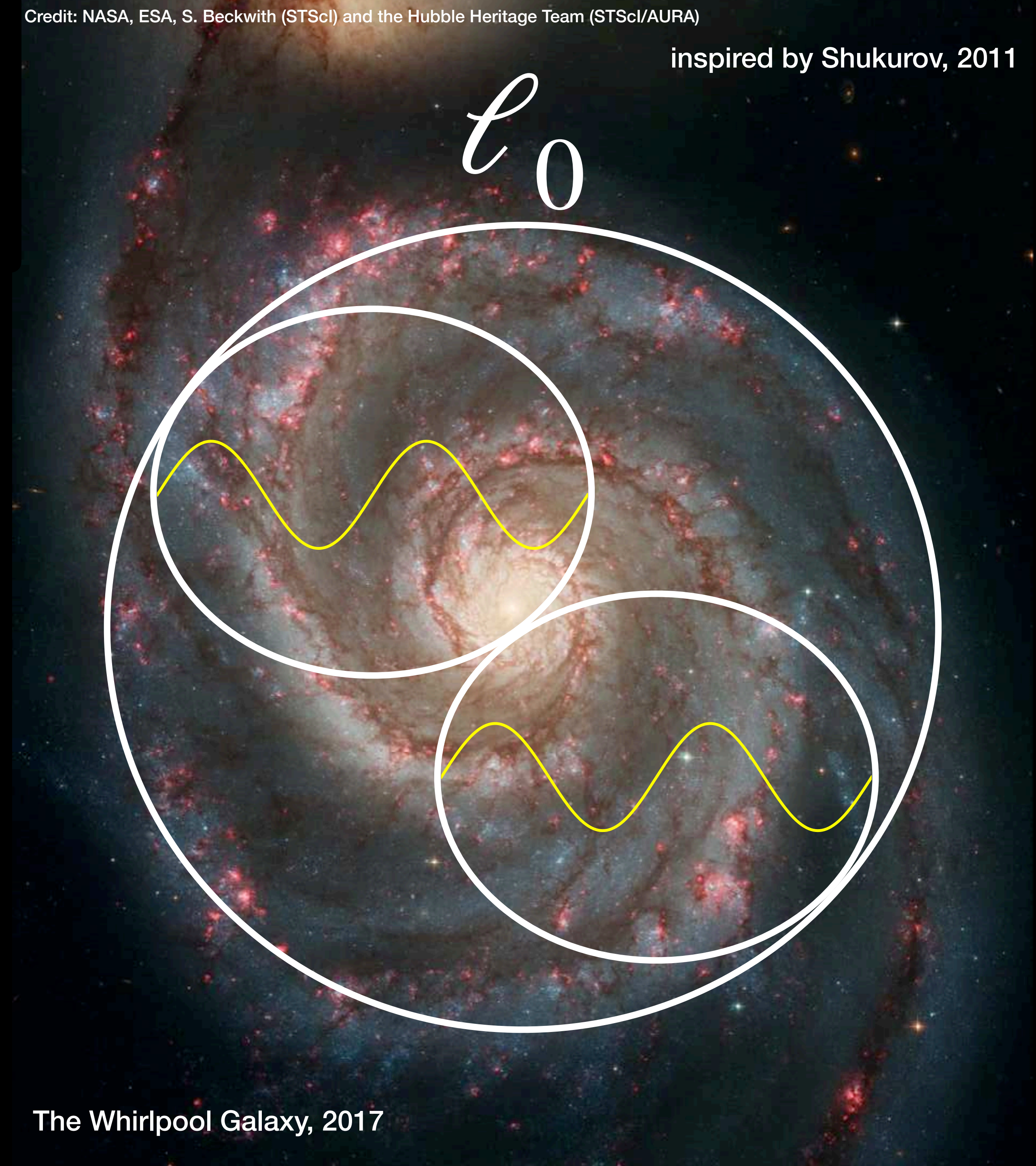
$$|\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u})|$$

dominates on large scales, ℓ

Creates new modes on

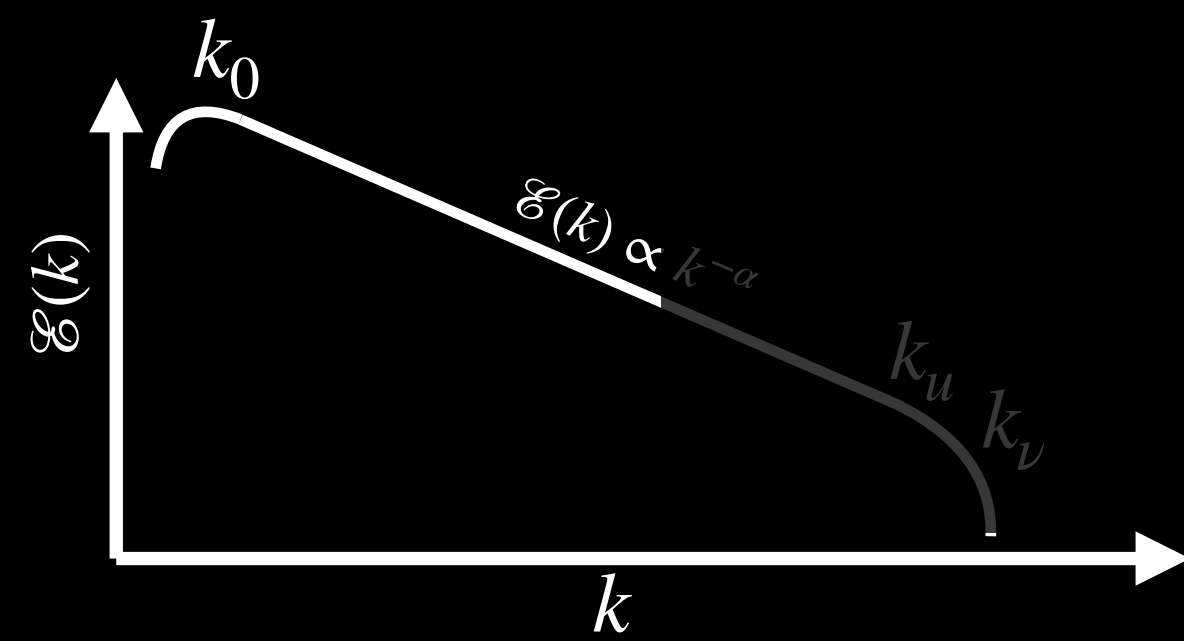
$$t_{\text{nl}} \sim L/U \sim \mathcal{O}(\text{Gyr})$$

$$\dot{\varepsilon} \sim u_0^3 / \ell_0$$



The Whirlpool Galaxy, 2017

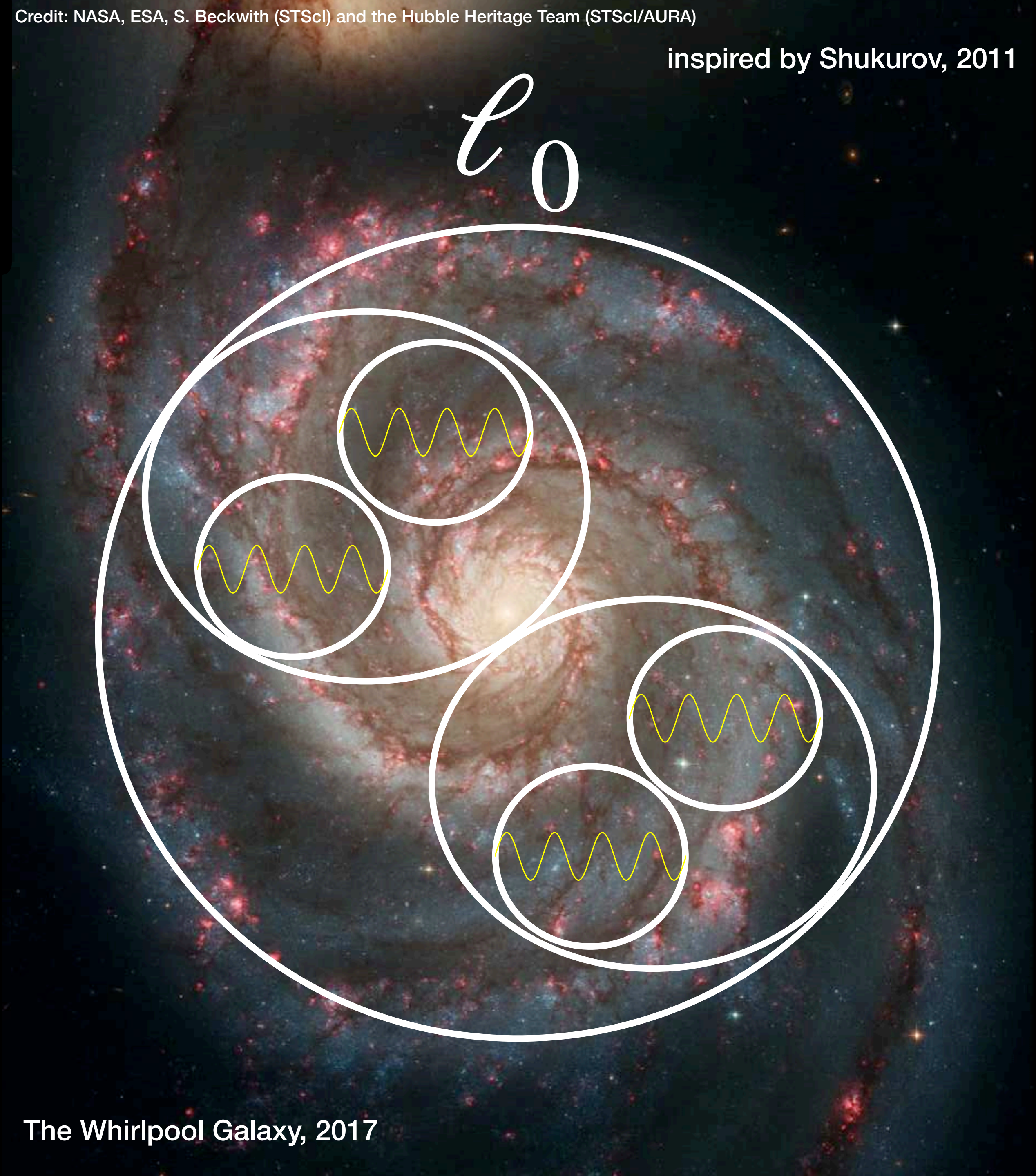
Turbulence Cascade



inside of the cascade

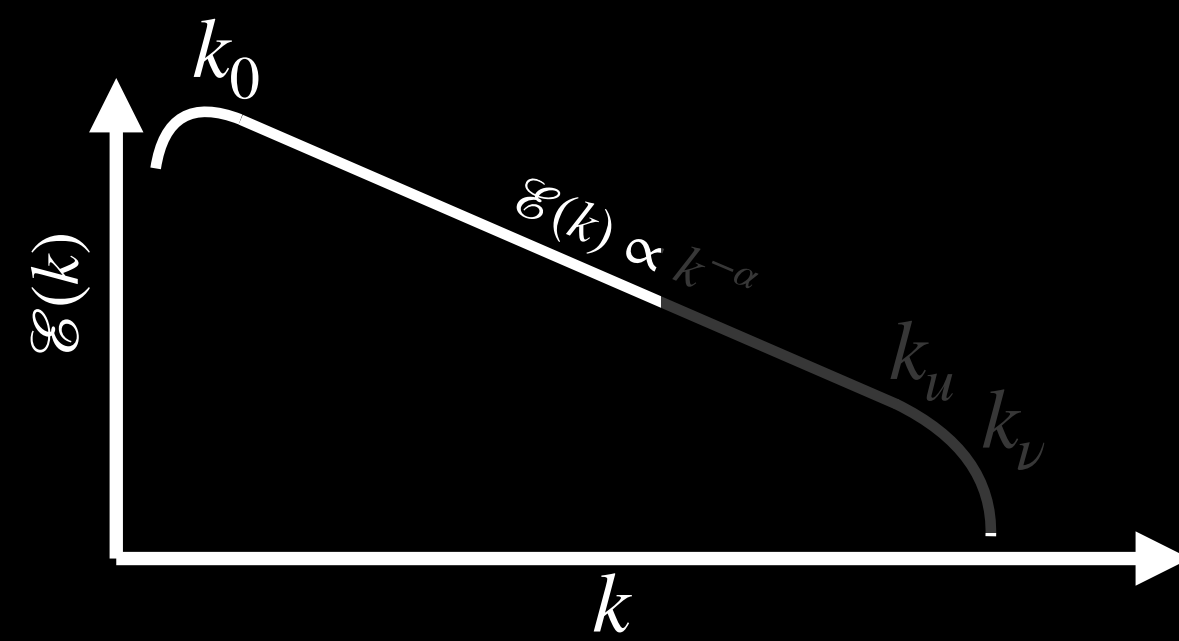
Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

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The Whirlpool Galaxy, 2017

Turbulence Cascade

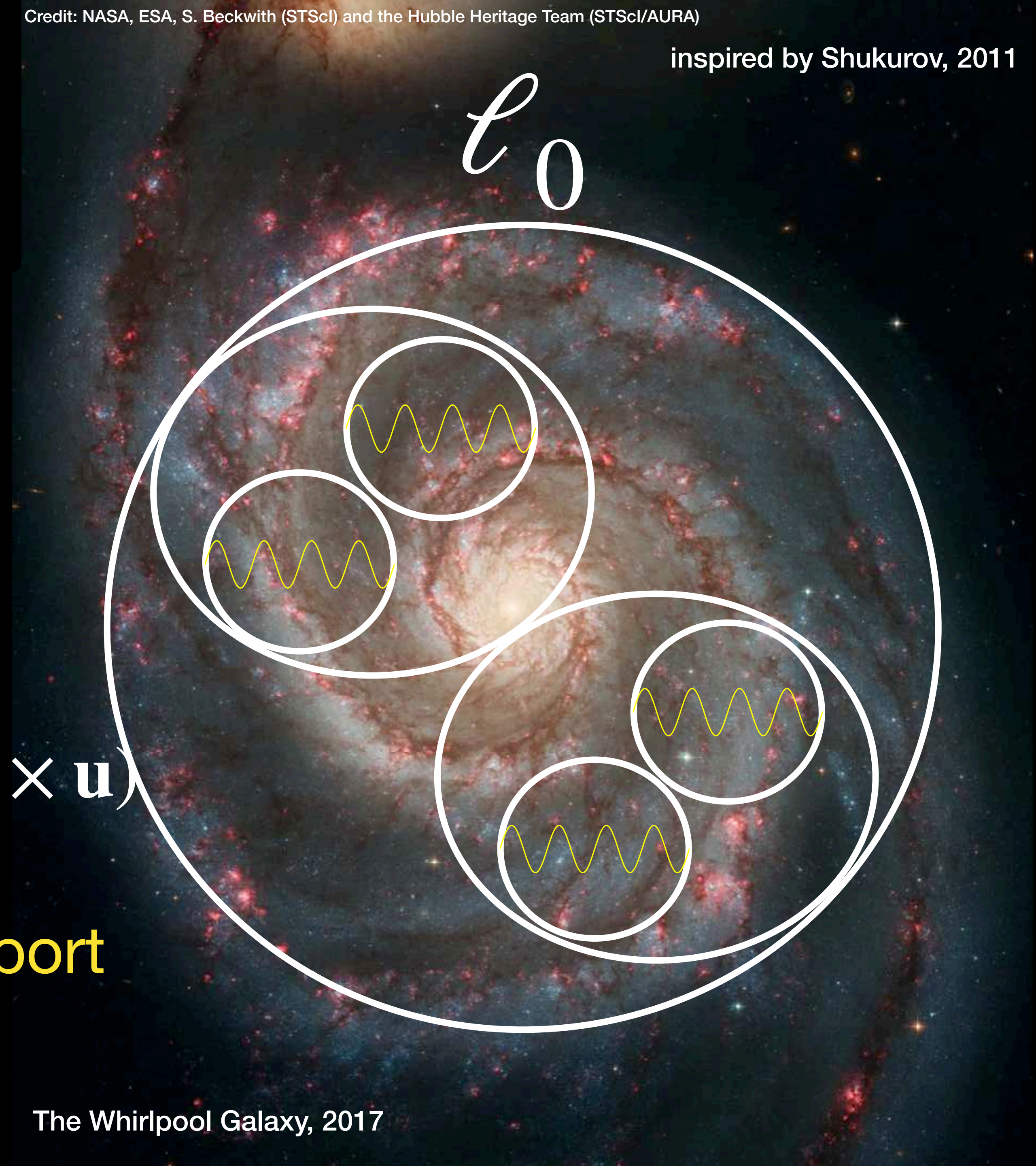


inside of the cascade

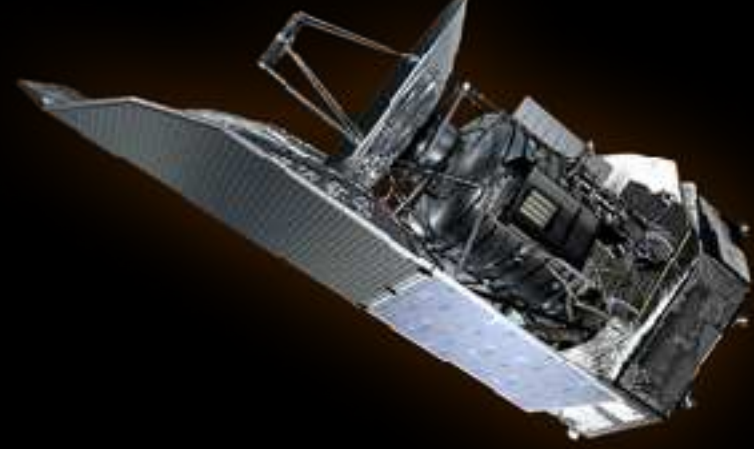
The quadratic nonlinear term

$$\nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \underbrace{1/2 \nabla u^2}_{\text{turbulent pressure}} - \mathbf{u} \times (\nabla \times \mathbf{u})$$

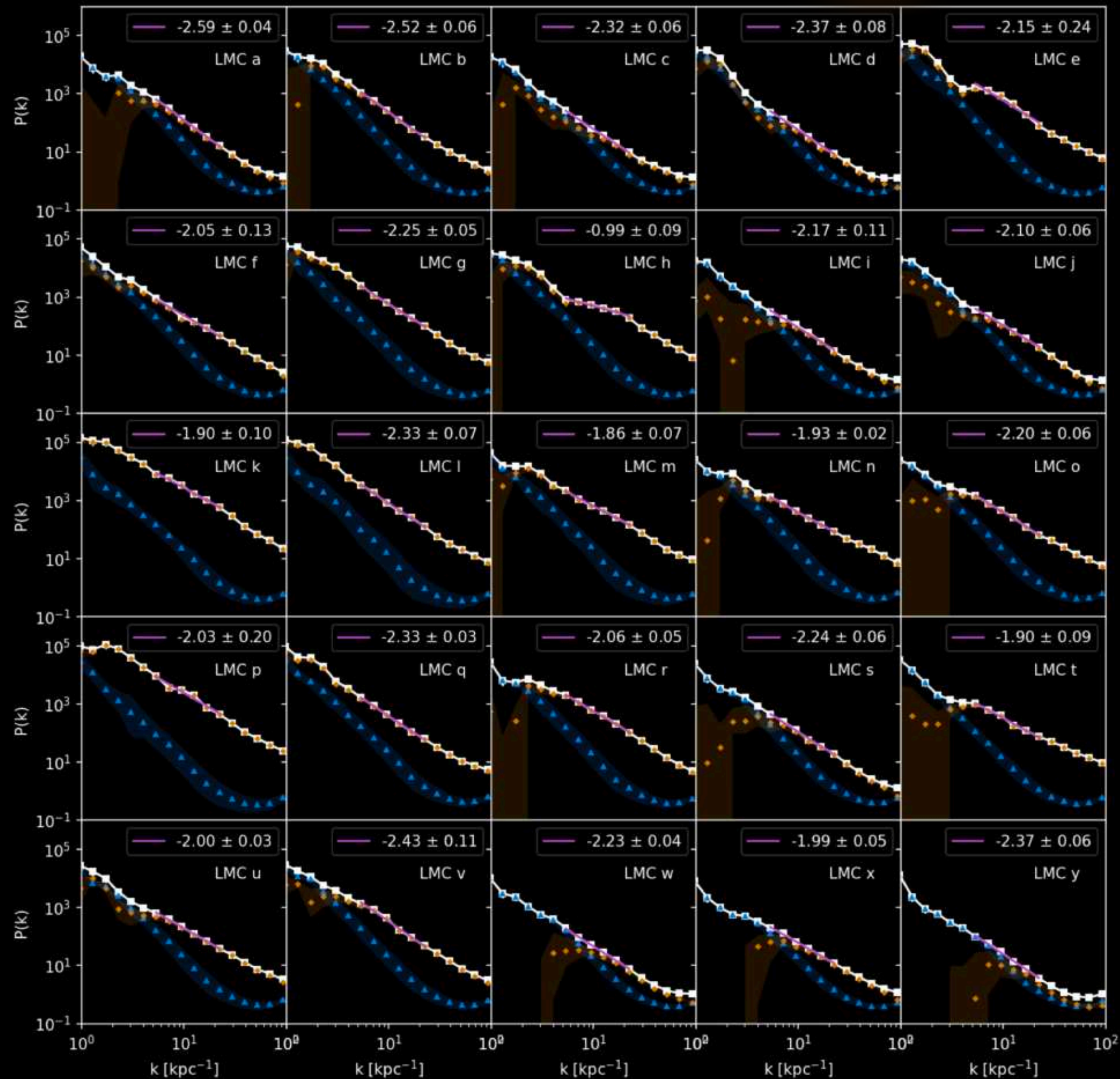
Turbulent pressure = turbulent support



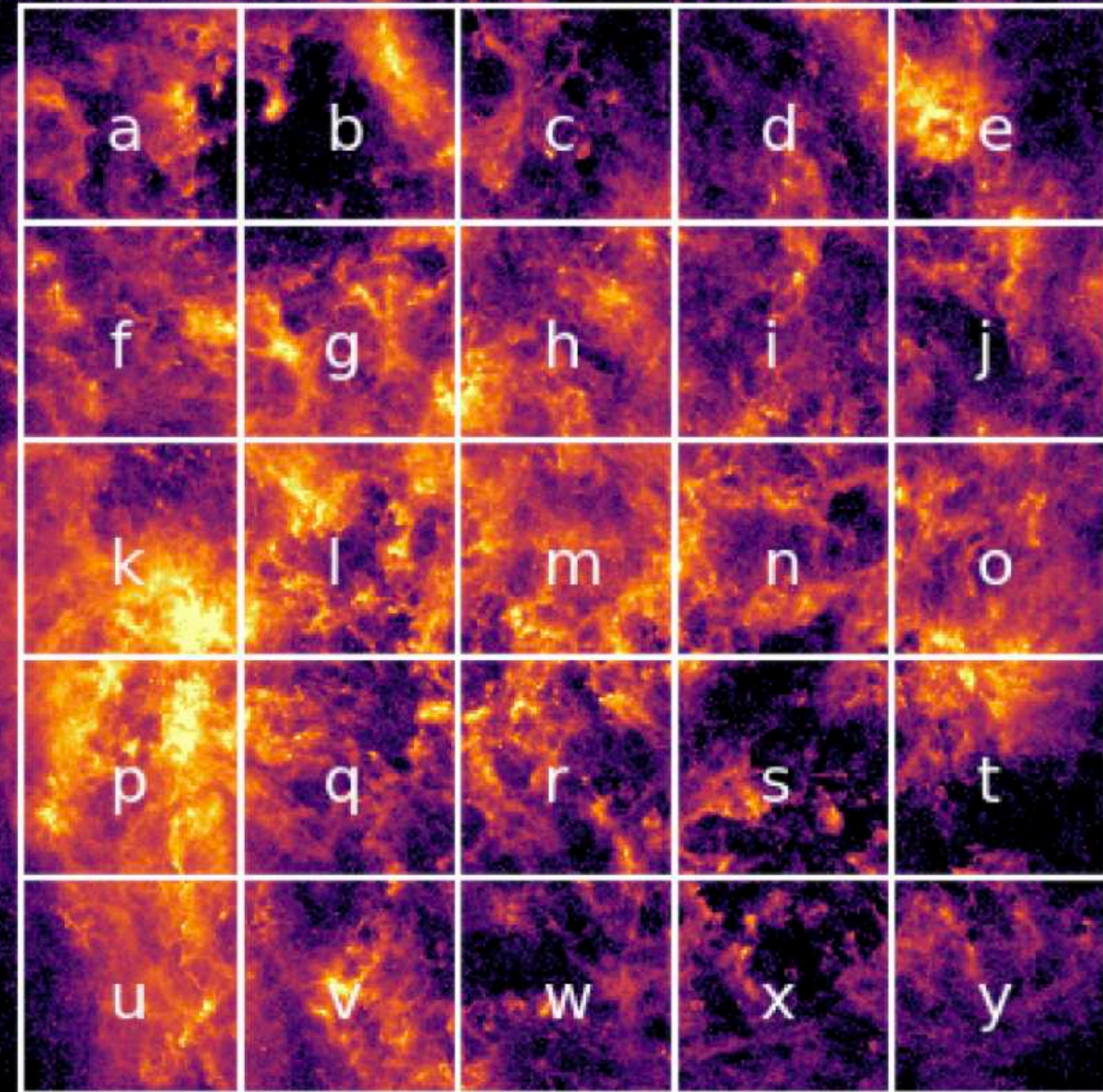
Turbulence



LMC: $500\mu\text{m}$, *Herschel* (processing Gordon et al. 2014)



1 kpc

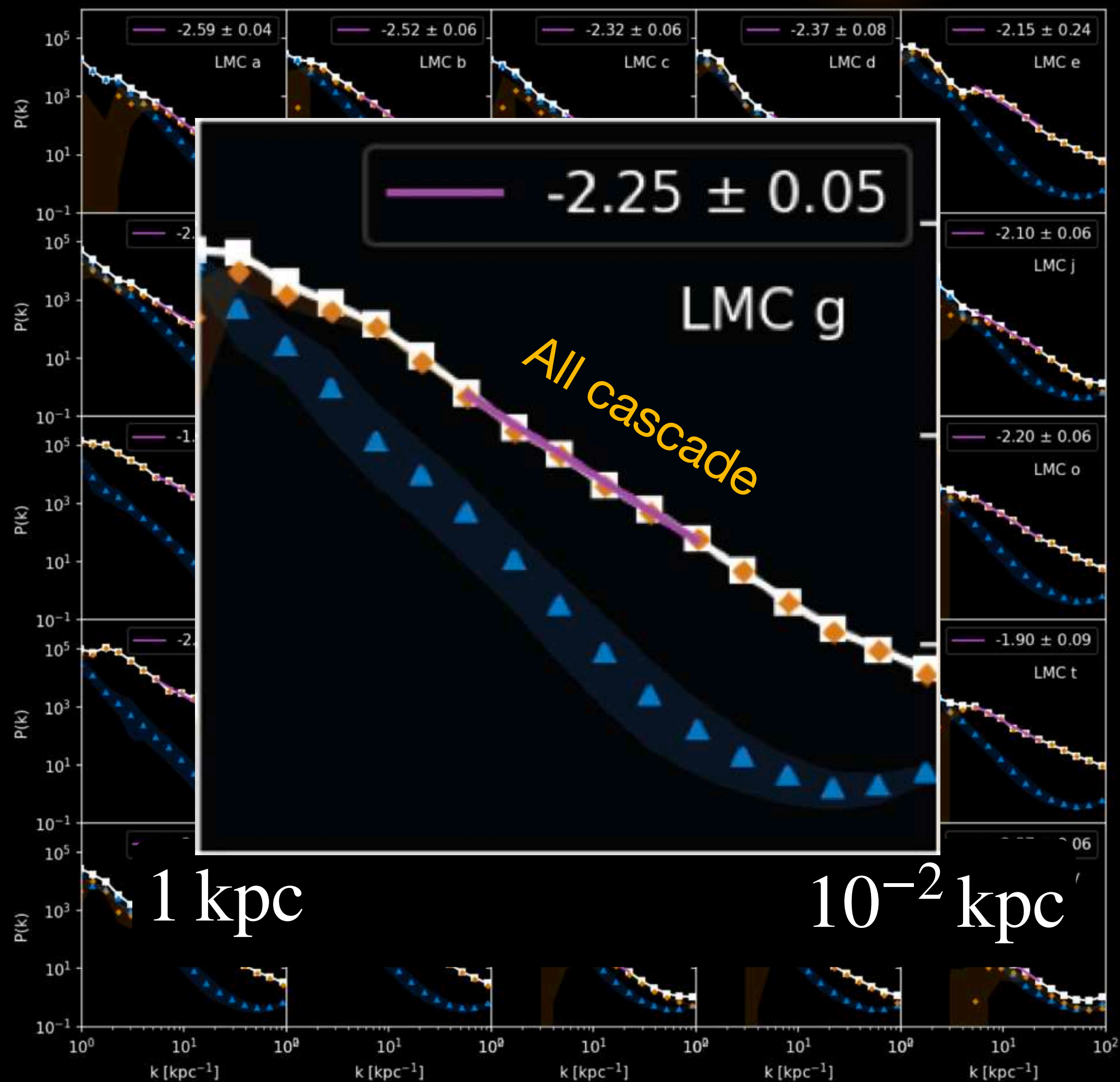


Colman, et al., 2022, MNRAS

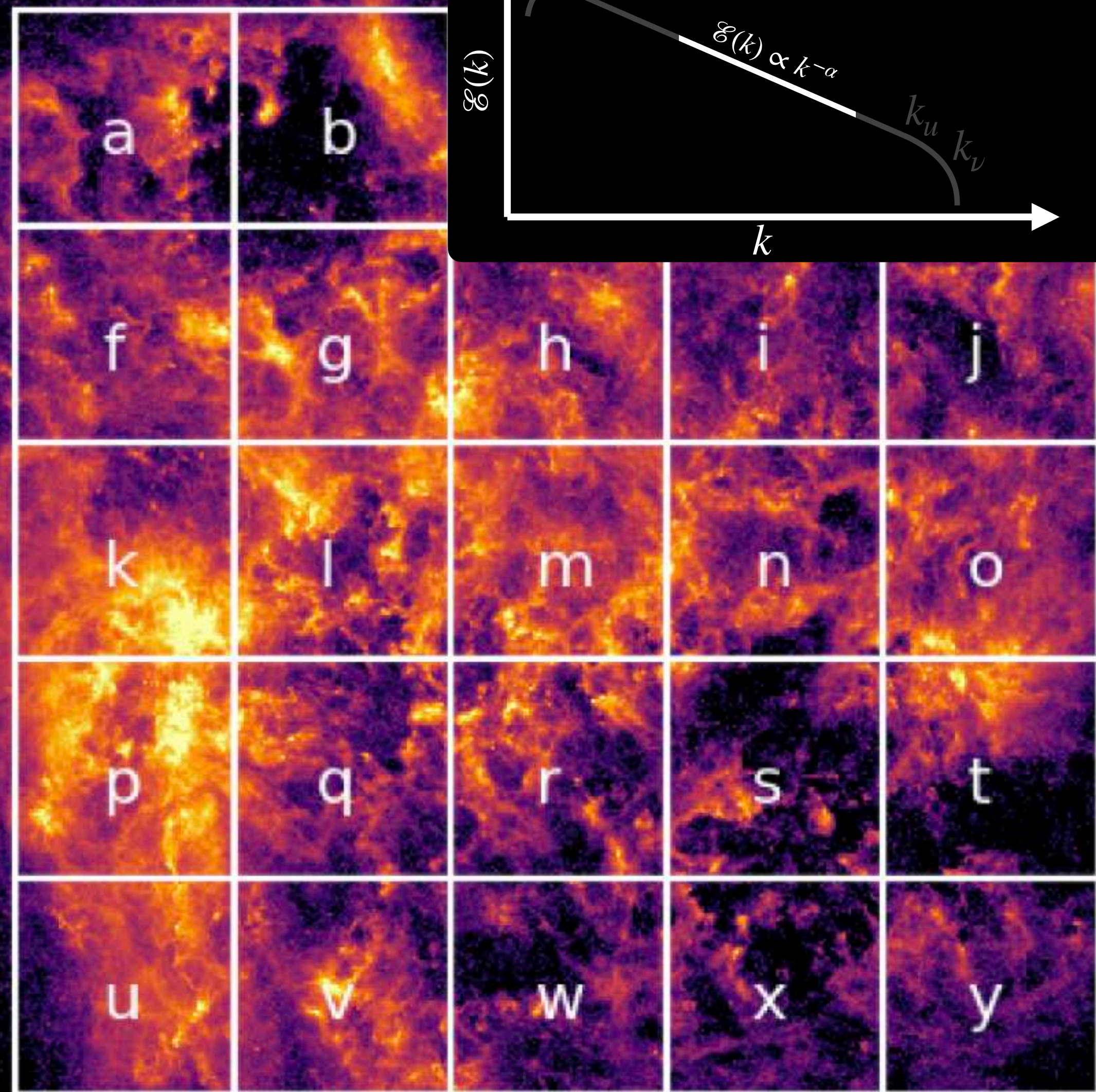
Turbulence



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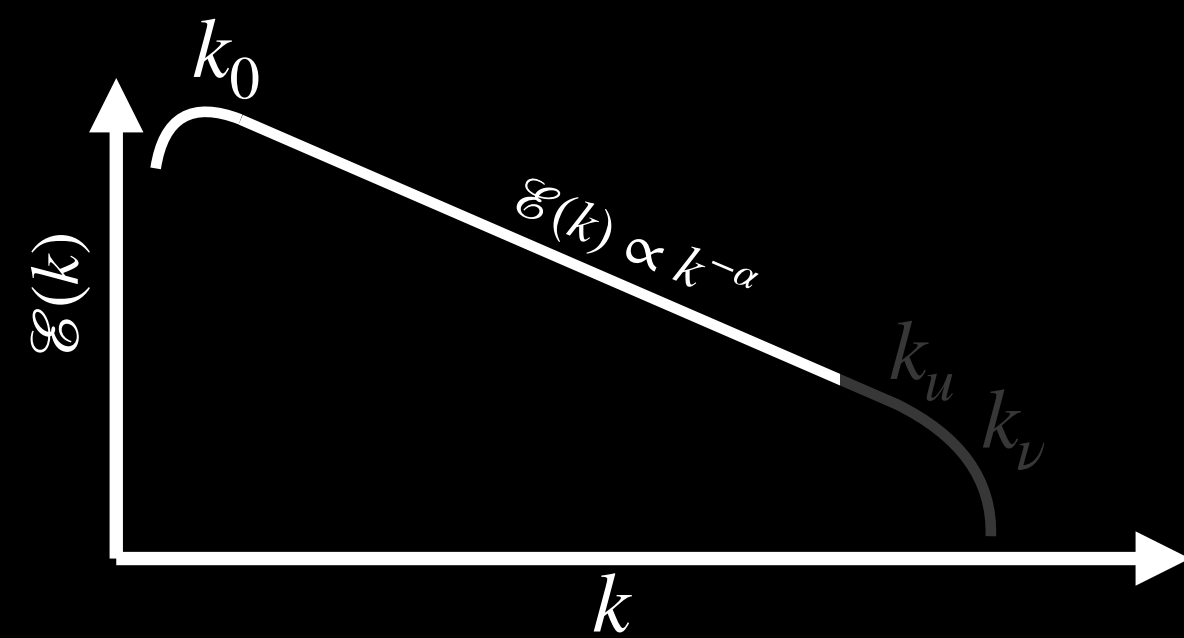


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Turbulence Cascade

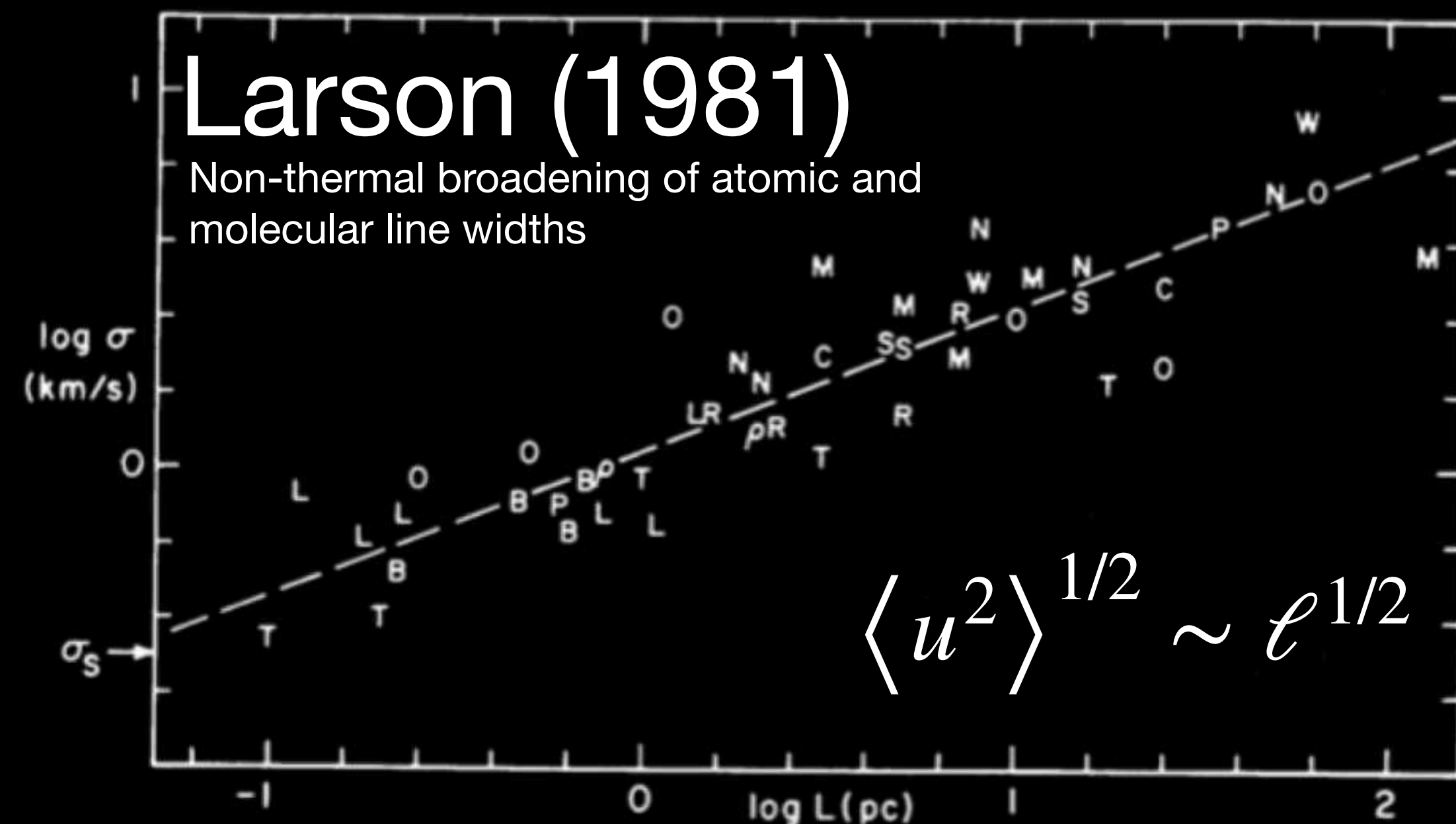


Credit: NASA, ESA, S. Beckwith (STScI) and the Hubble Heritage Team (STScI/AURA)

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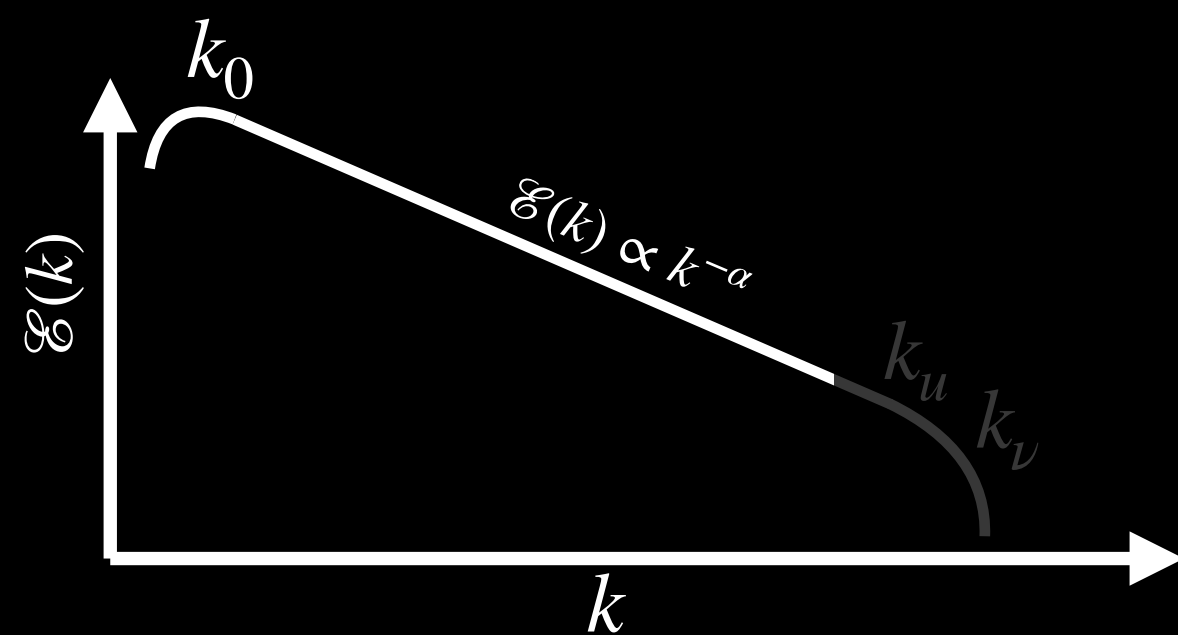
Reynolds number is shrinking

$$\text{Re} \sim \langle u^2 \rangle^{1/2} \ell / \nu \sim \ell^{3/2} / \nu$$



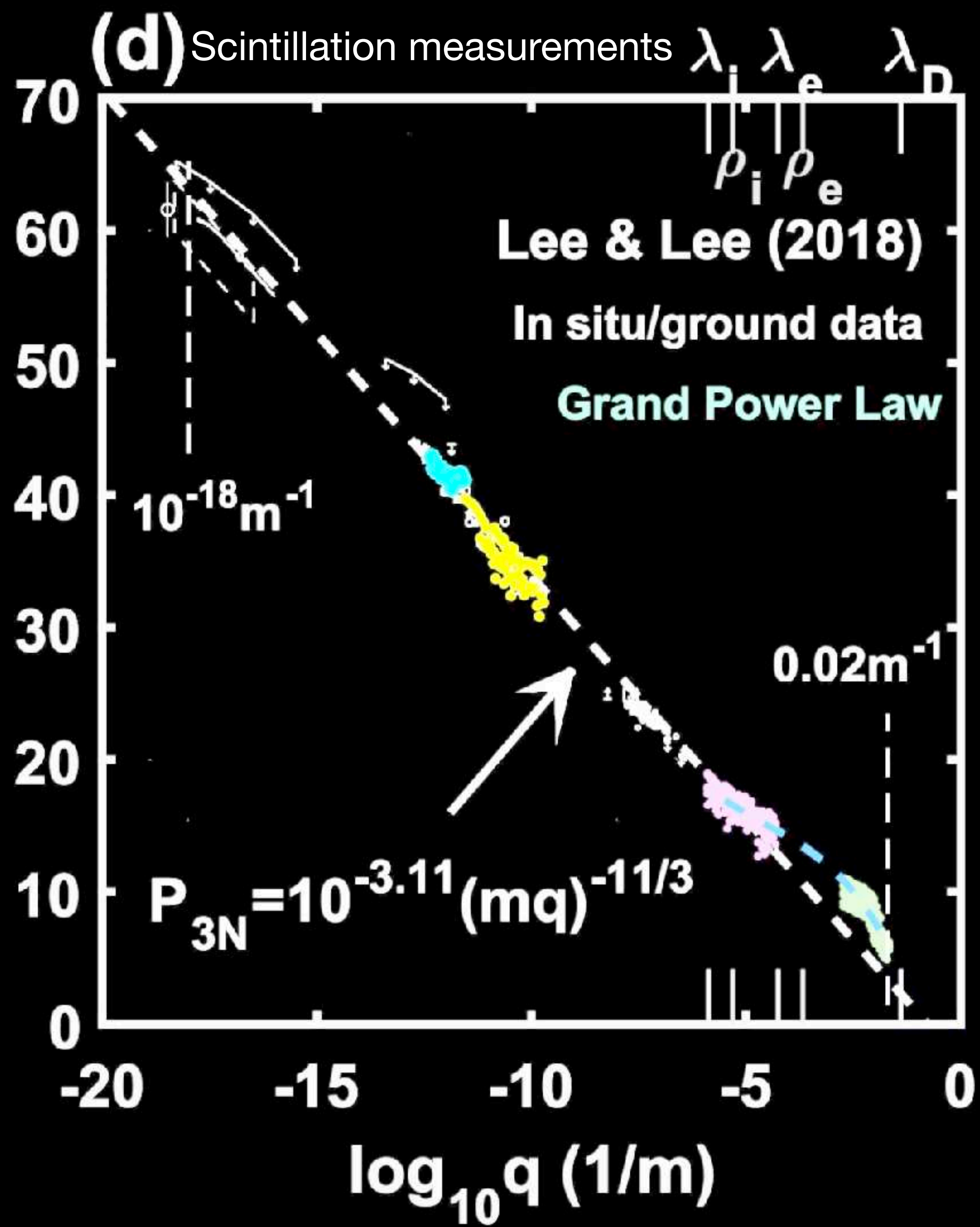
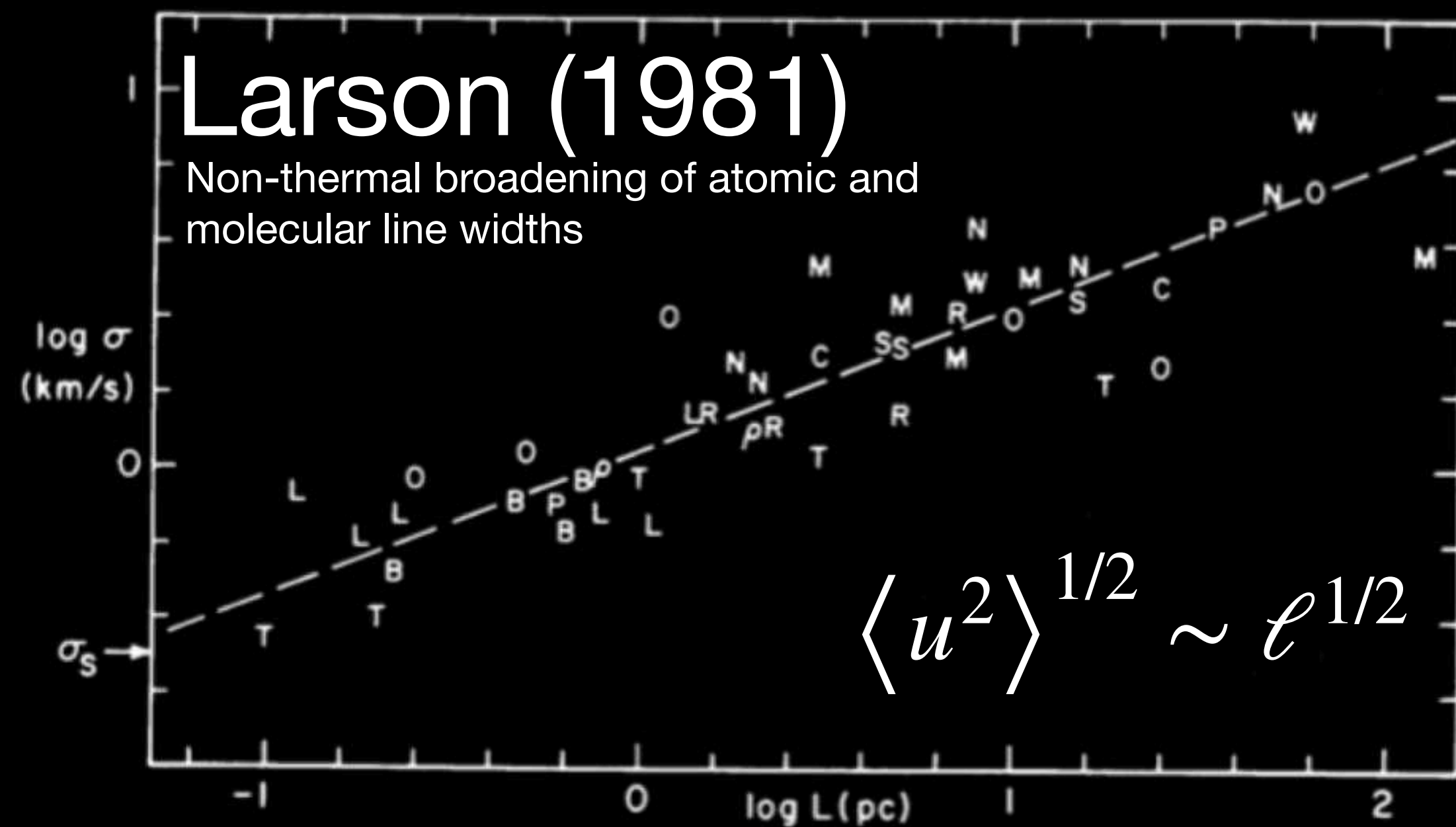
The Whirlpool Galaxy, 2017

Turbulence Cascade

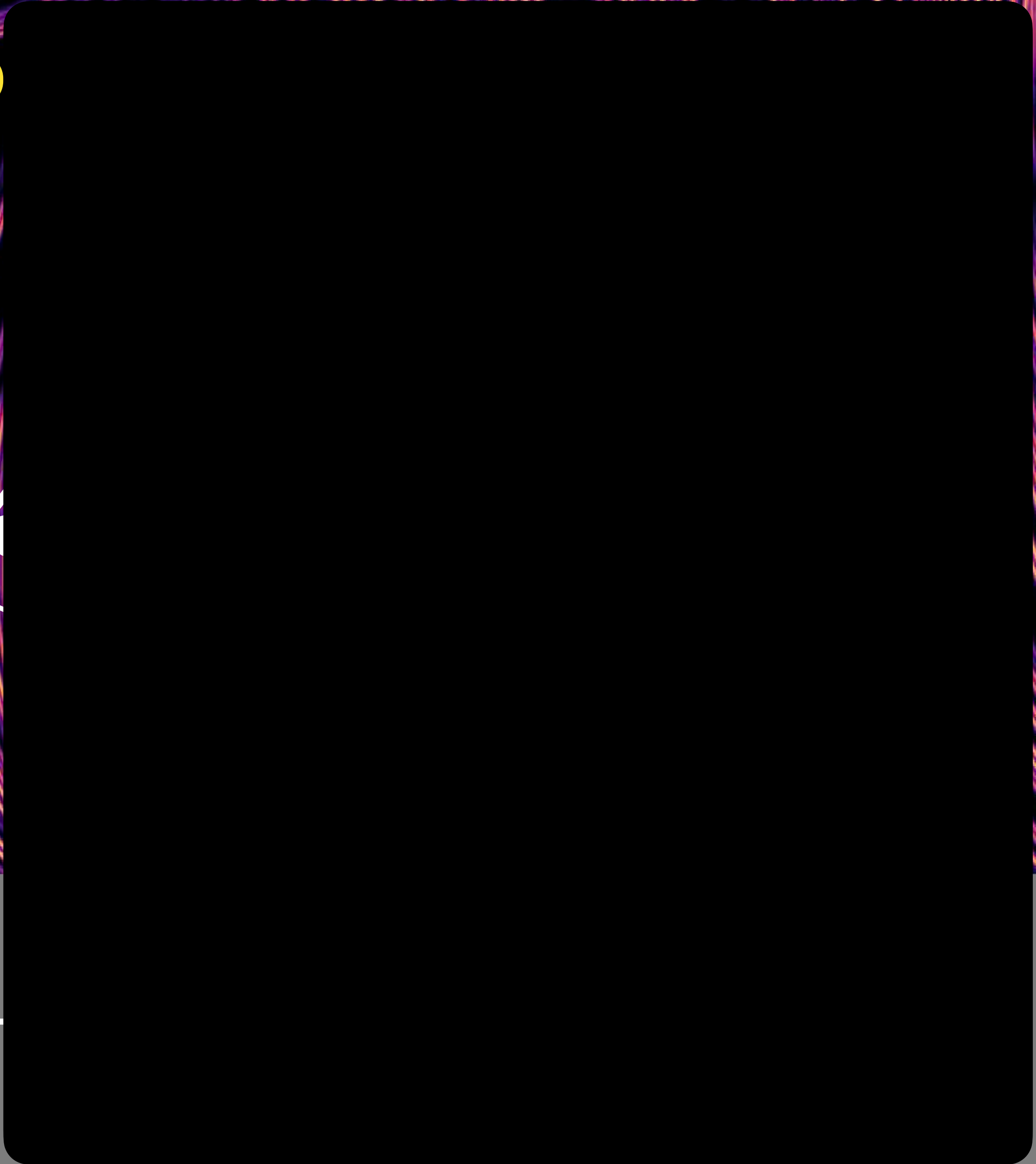
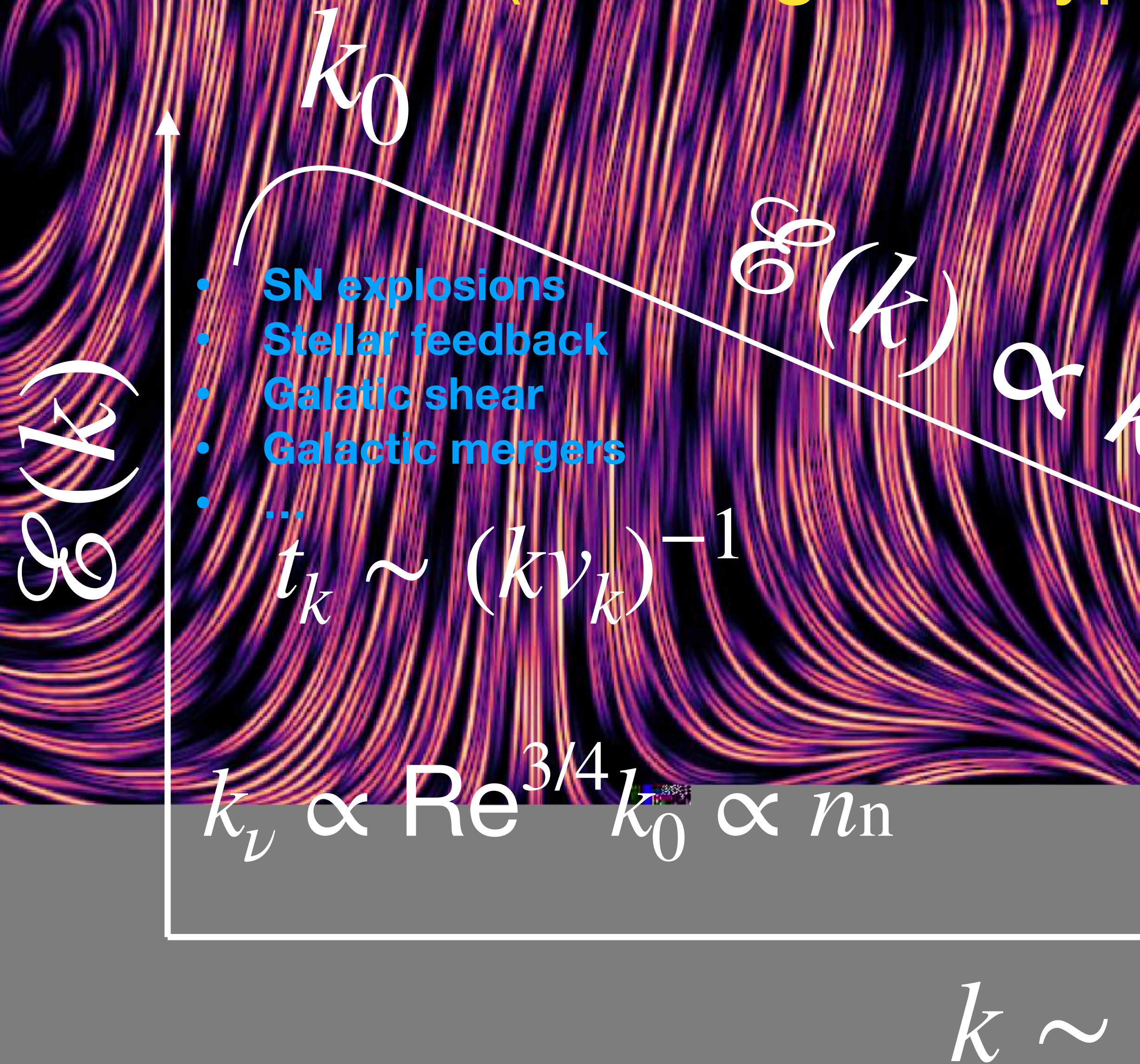


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The (Kolmogorov-type



Why should astrophysicists care — cosmic ray transport?

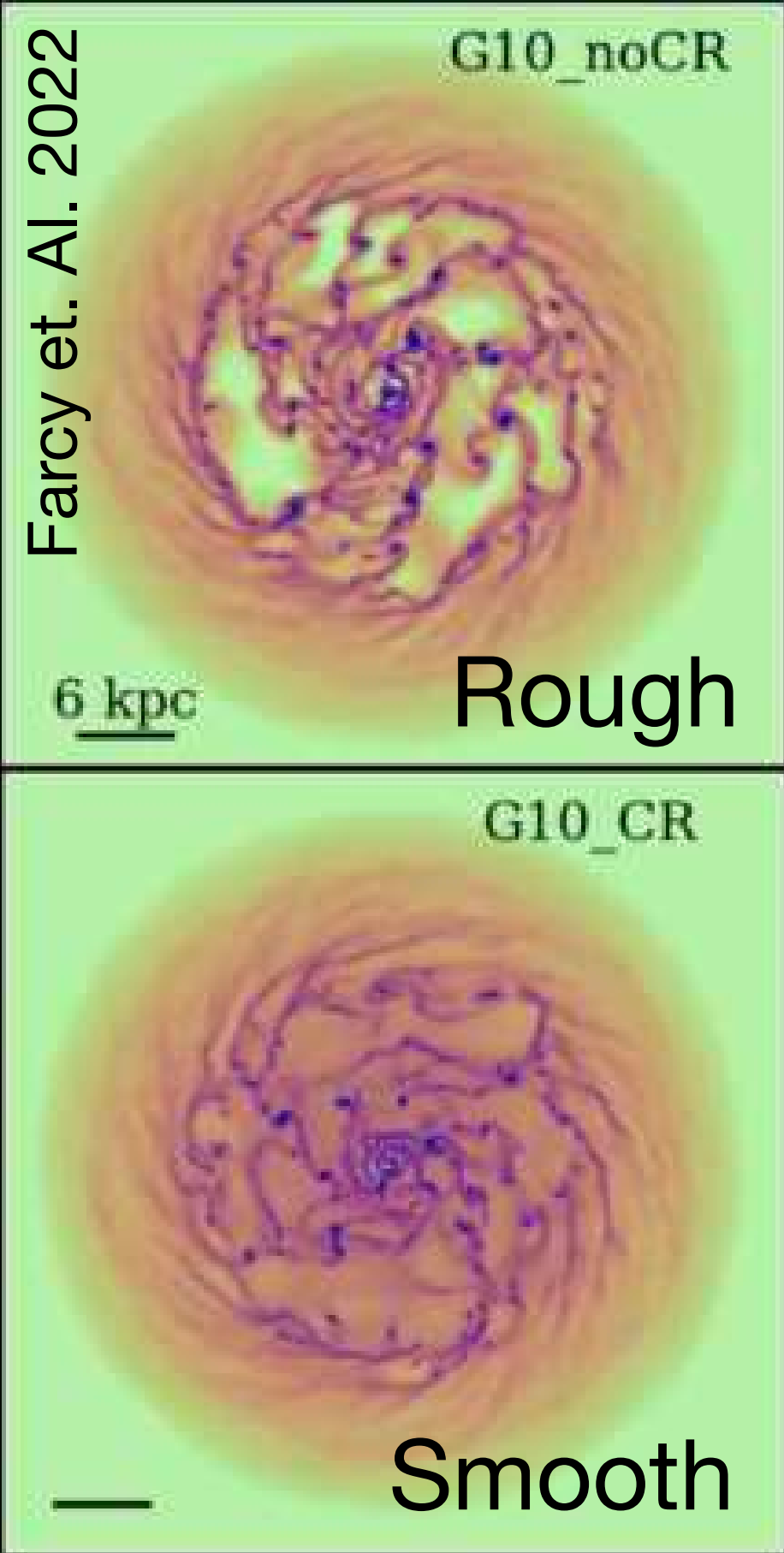
Testing Physical Models for Cosmic Ray Transport Coefficients on Galactic Scales: Self-Confinement and Extrinsic Turbulence at $\sim$ GeV Energies

Philip F. Hopkins¹, Jonathan Squire², T. K. Chan^{3,4}, Eliot Quataert⁵,
Suoqing Ji¹, Dušan Kereš², Claude-André Faucher-Giguère⁶

CRs add extra pressure support + smooth plasma gradients

Provides source of diffusion on large-scales

Determines the properties of turbulence



Sampson, Beattie et al. (2023, MNRAS). *Turbulent diffusion of streaming cosmic rays in compressible, partially ionised plasma*

Beattie et al. (2023, Frontiers). *Ion Alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays*

The energy cascade — fundamental for CR transport

Two great insights from Kolmogorov

- Constant energy flux between modes
- No magnetic field
- No inhomogeneities
- No density fluctuations
- Isotropic

$$\varepsilon \sim u_\ell^3 / \ell \sim \text{const.}$$

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$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

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$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

$$u_\ell \sim (\varepsilon \ell)^{1/3} \iff u^2(k) \sim k^{-5/3} dk$$

Two great insights from Kolmogorov

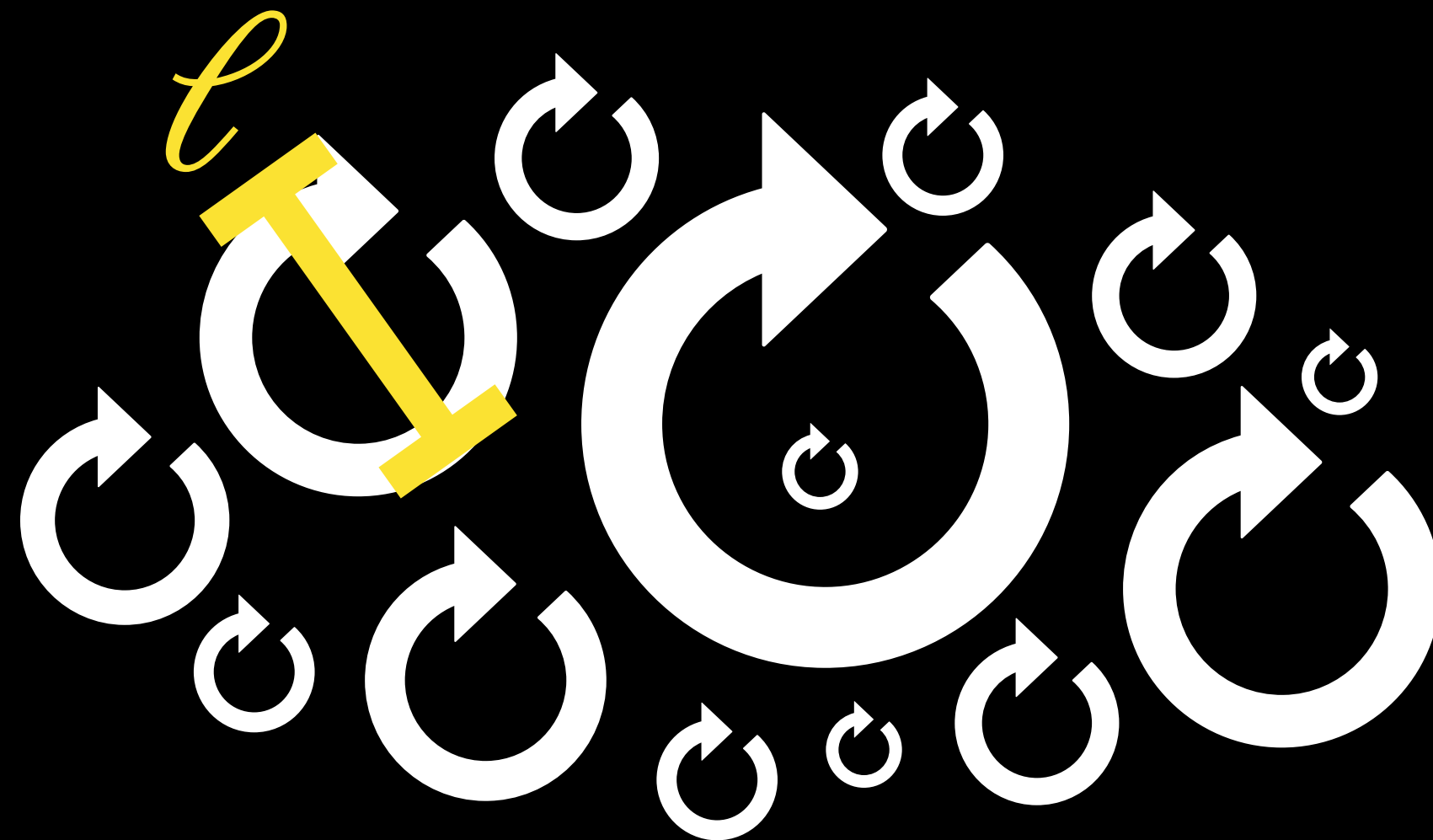
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$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

“structure function”

$$u(\ell) = \langle |u(\mathbf{x}) - u(\mathbf{x} + \ell) \cdot \ell| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

ℓ picks out an eddy



Two great insights from Kolmogorov

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$$u_\ell \sim (\varepsilon \ell)^{1/3}$$

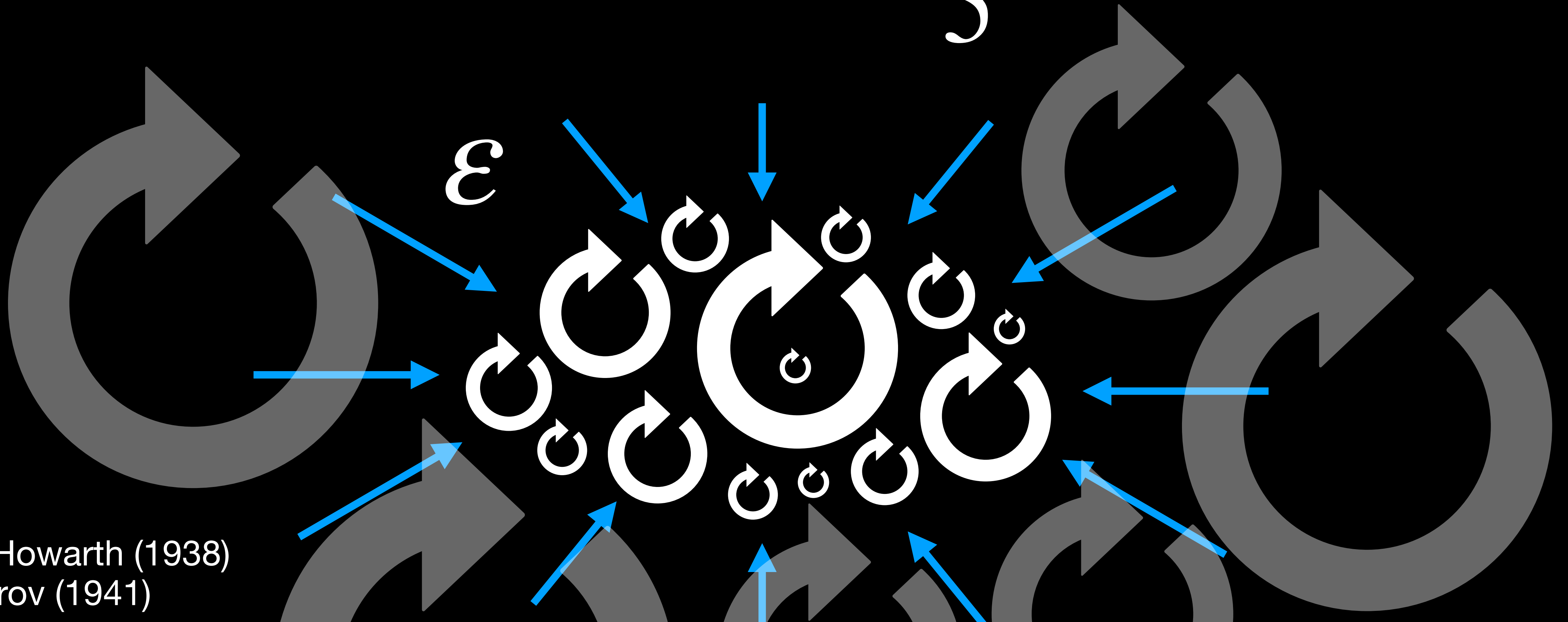
$$u(\ell) = \langle |u(\mathbf{x}) - u(\mathbf{x} + \ell) \cdot \ell| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

$$u(\ell) = \langle |\delta u_L| \rangle_{\mathbf{x}} \sim \ell^{1/3}$$

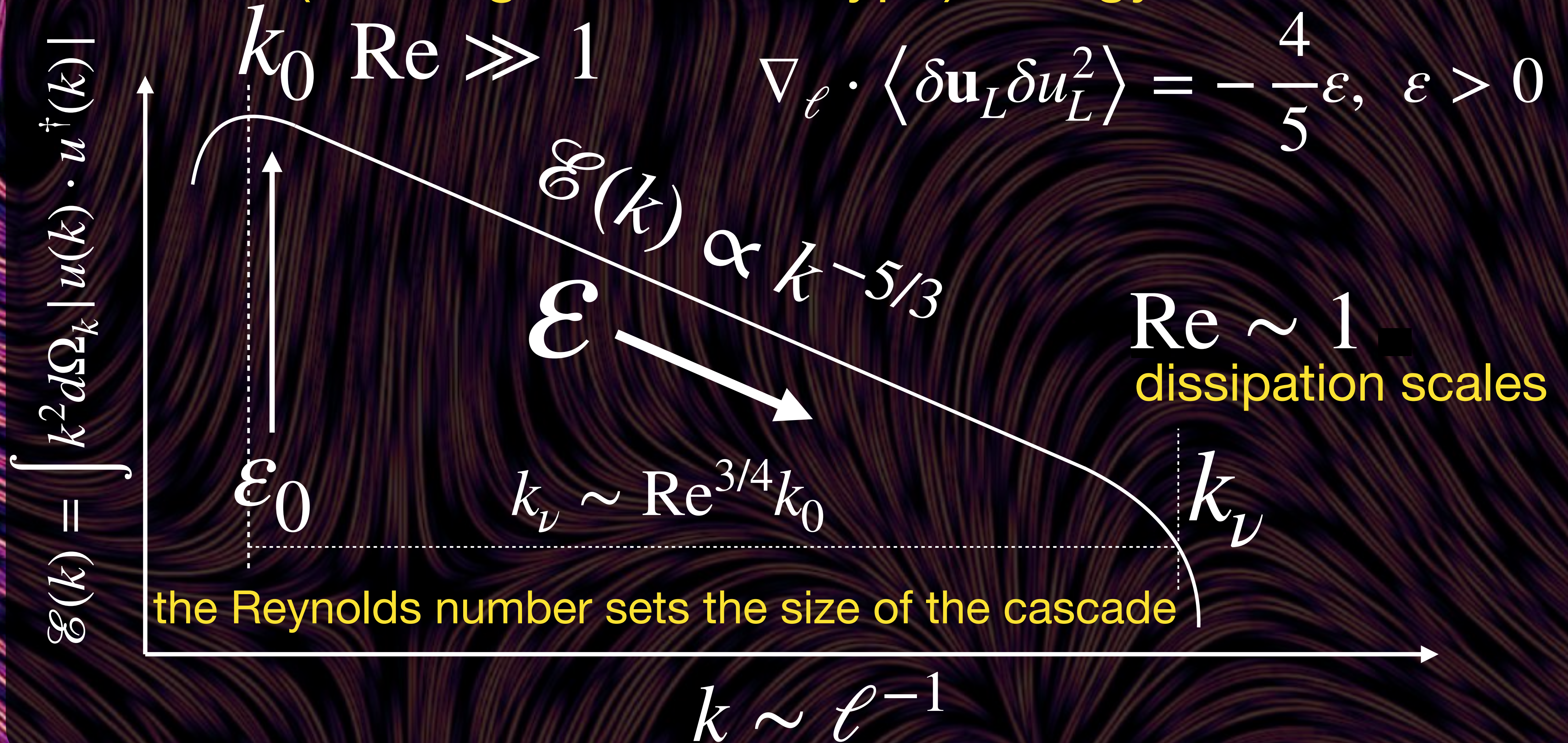
An exact relation for turbulence

Kolmogorov's 4/5 law

$$\nabla_{\ell} \cdot \langle \delta \mathbf{u}_L \delta u_L^2 \rangle = -\frac{4}{5} \varepsilon, \quad \varepsilon > 0$$



The (Kolmogorov, 1941-type) energy cascade



Fact: the galaxy that we reside in is in a state of highly-compressible fluid turbulence

$$E_{\text{SNe}} \approx 10^{51} \text{ erg}$$

$$\gamma_{\text{SNe}} \approx 0.01 - 0.03 \text{ yr}^{-1} \quad \text{Diehl + (2016)}$$

$$\dot{E}_{\text{SNe}} \approx E_{\text{SNe}} \gamma_{\text{SNe}} \approx 10^{41} \text{ erg s}^{-1}$$

$$E_{\text{turb}} \approx 10^{54} \text{ erg}$$

$$t_{\text{turb}} \approx \ell_0 / \langle u^2 \rangle^{1/2} \approx 10^7 \text{ yr} \quad \text{Ferriere + (2020)}$$

$$\dot{E}_{\text{turb}} \approx E_{\text{turb}} / t_{\text{turb}} \approx 10^{39} \text{ erg s}^{-1}$$

$$\dot{E}_{\text{turb}} / \dot{E}_{\text{SNe}} \approx 10^{-2}$$

CC-supernovae alone provide enough energy flux to fuel a continuously driven cascade

Connor, **Beattie** + (2025, ApJ) Cascading from the winds to the disk: universality of supernovae-driven turbulence in different galactic ISMs

The (Kolmogorov, 1941-type) energy cascade

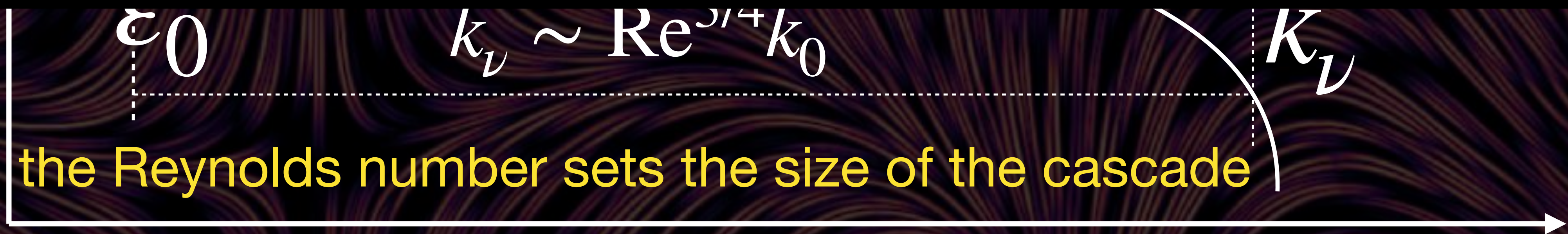
$$k_0 \text{ Re} \gg 1$$

$$\nabla_{\ell} \cdot \langle \delta \mathbf{u}_L \delta u_L^2 \rangle = -\frac{4}{5} \varepsilon, \quad \varepsilon > 0$$

Does it resemble this
idea at all?

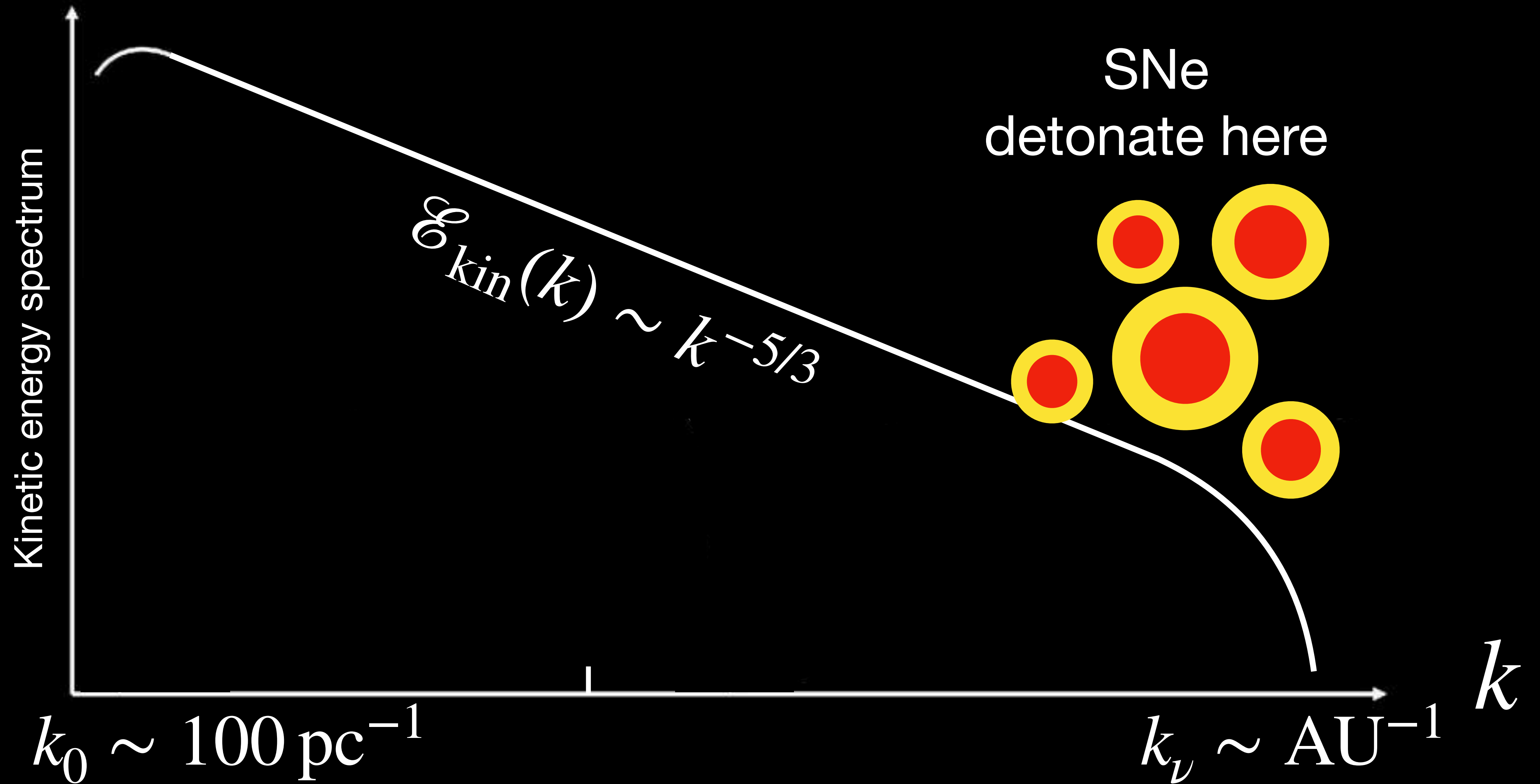
ales

$$\mathcal{E}(k) = \int k^2 d\Omega_k |u(k) \cdot u^{\dagger}(k)|$$

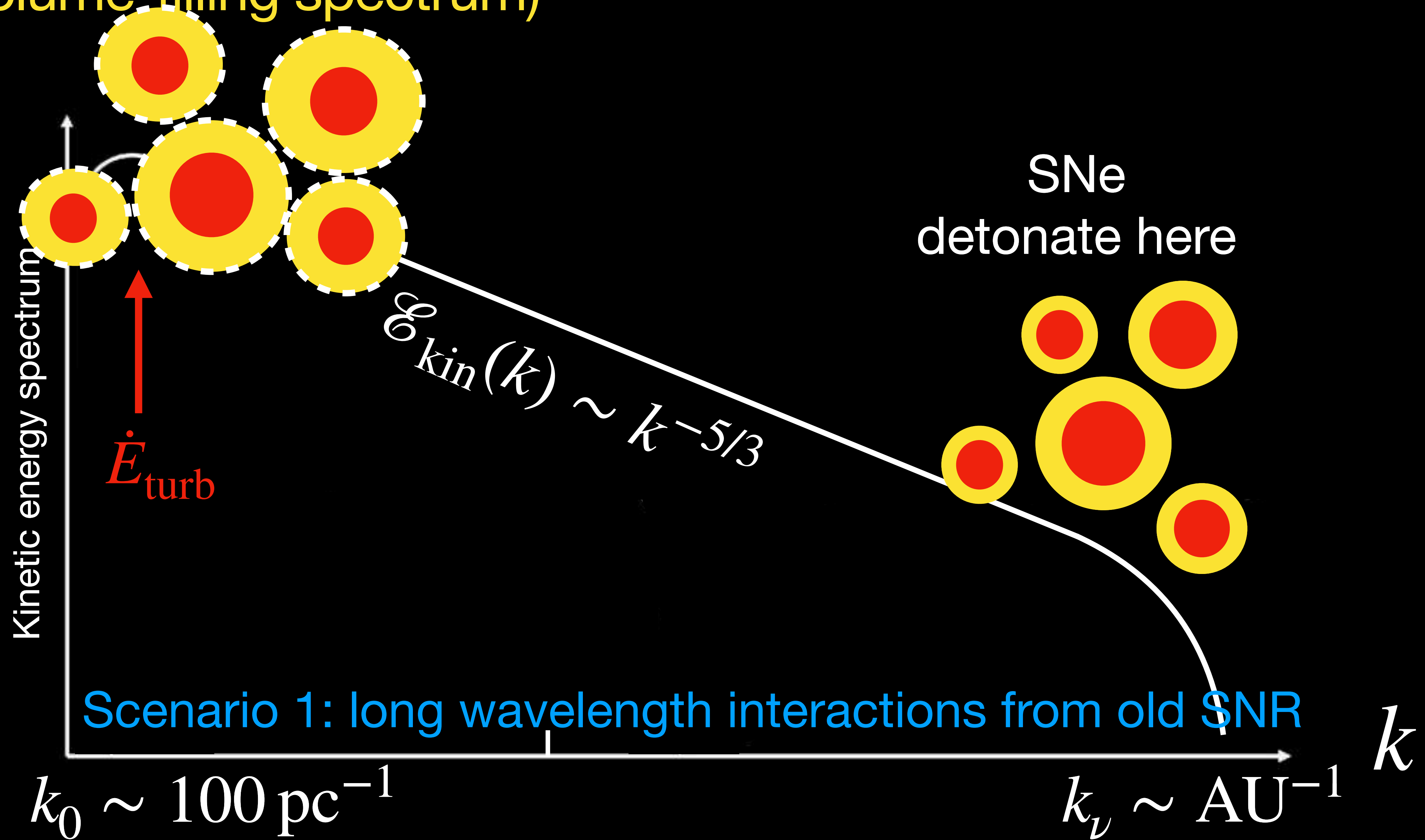


$$k \sim \ell^{-1}$$

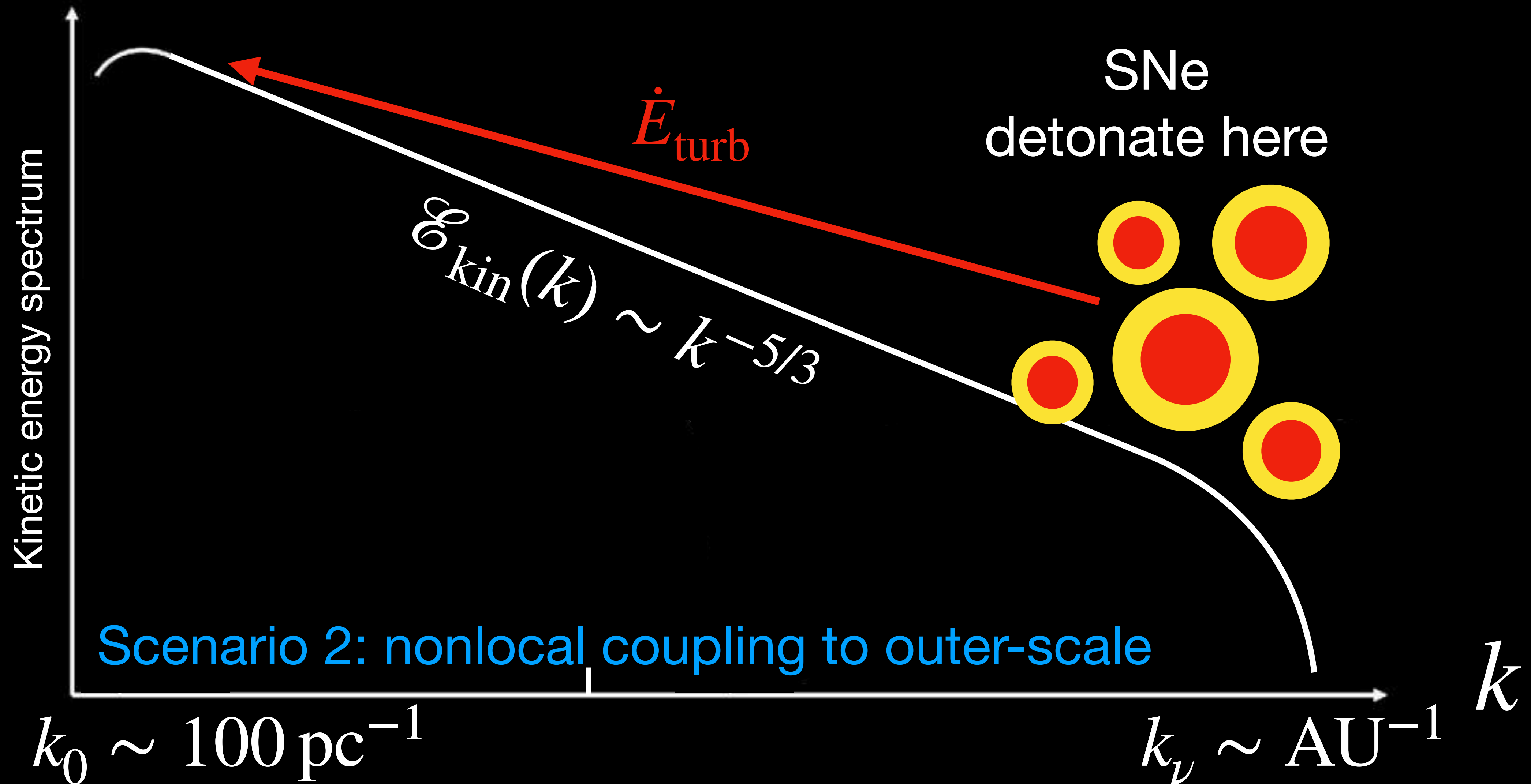
The naive ($\mathcal{M} \approx 2$) warm ionized medium spectrum
(the volume-filling spectrum)



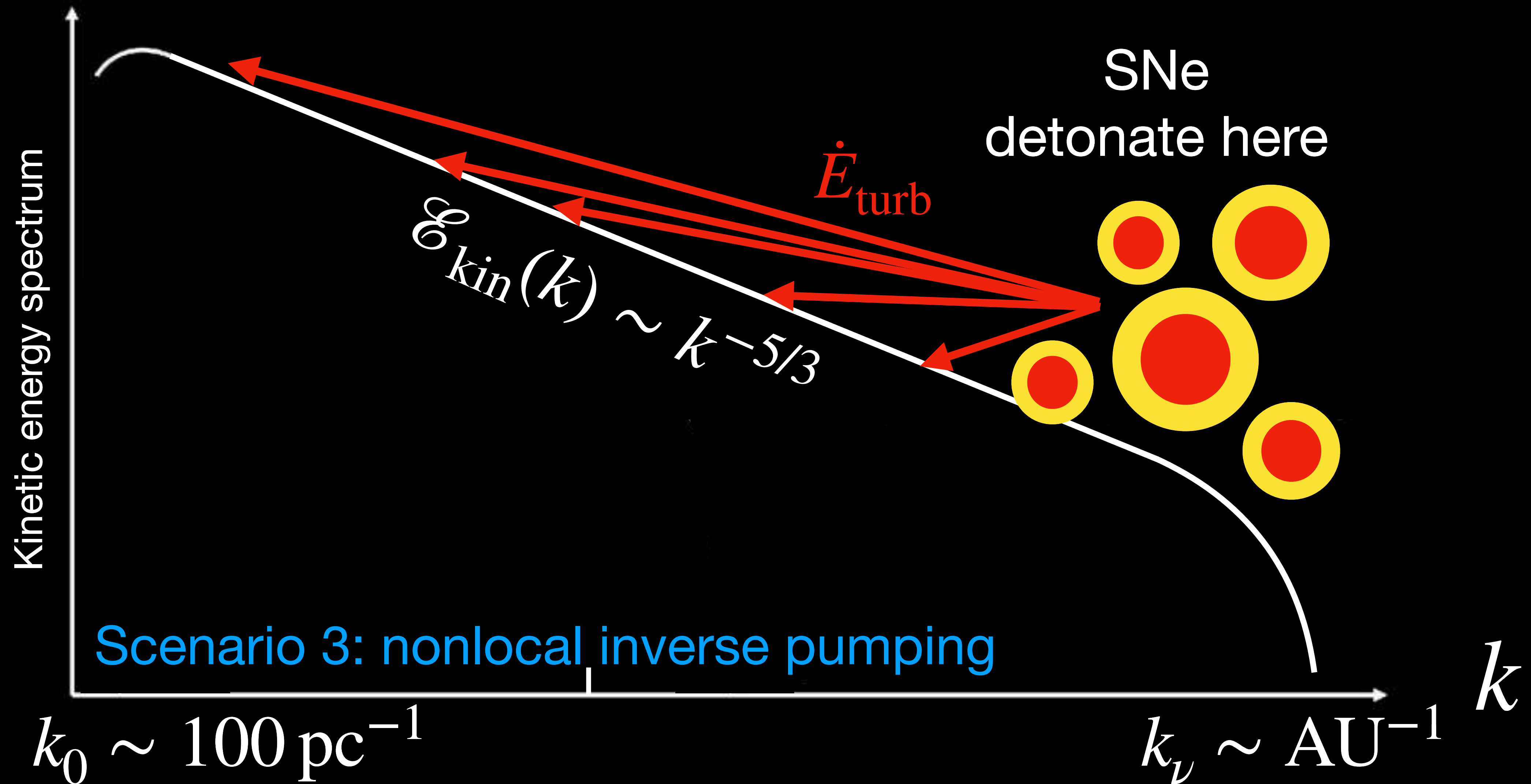
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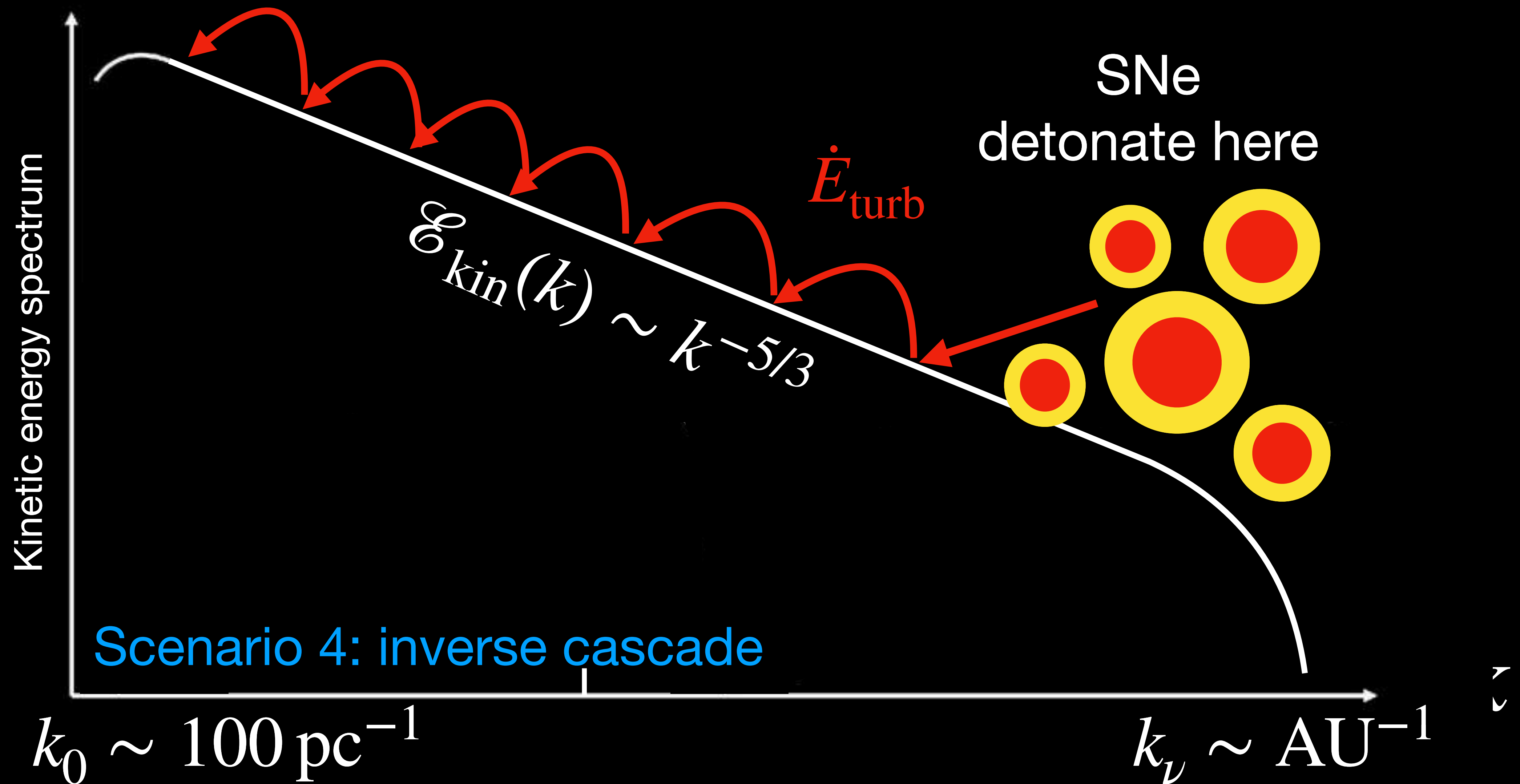
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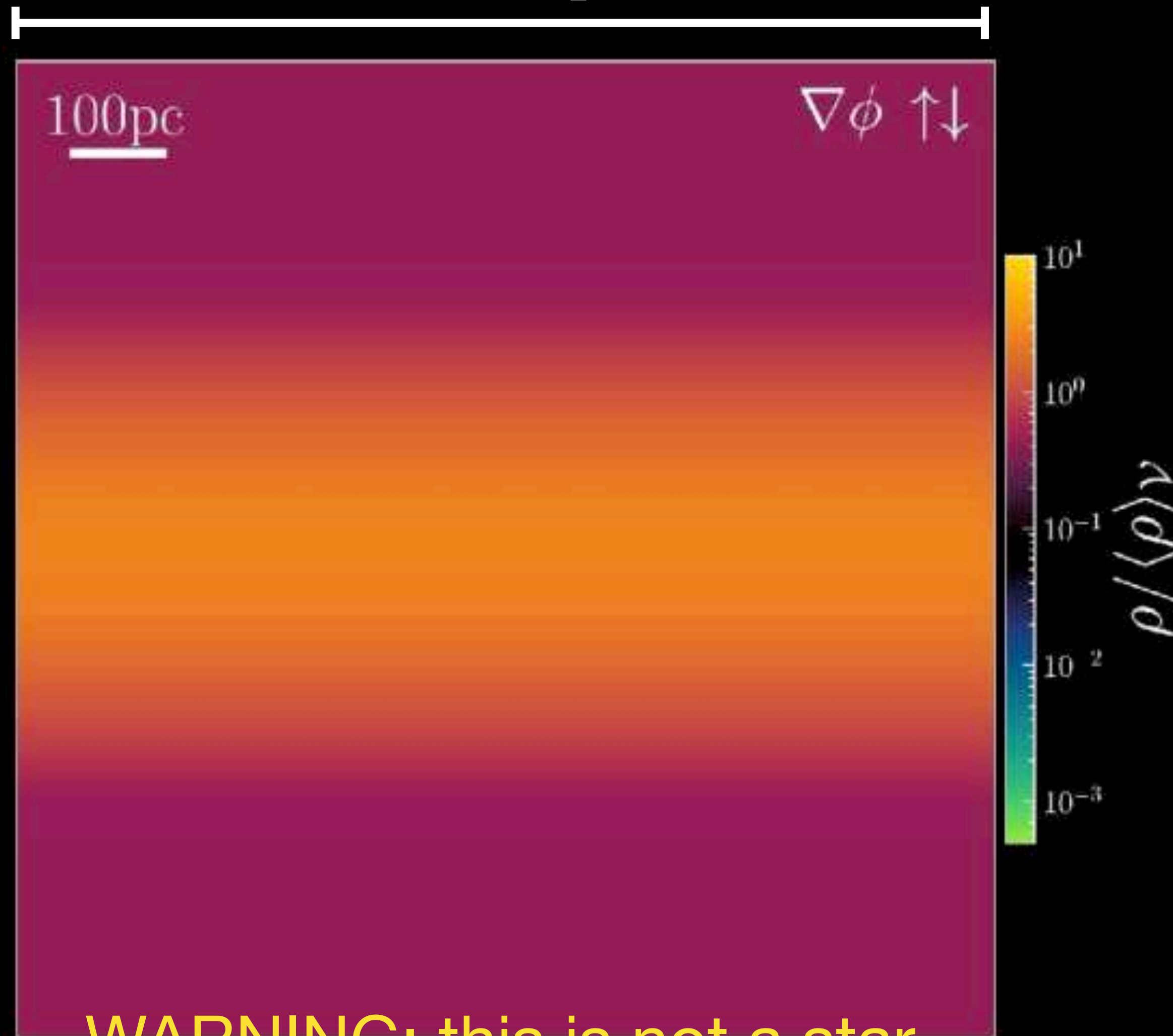
The simplest SNe-driven simulations possible

$L = 1 \text{ kpc}$

Martizzi+2016

RAMSES (Teyssier 2002)

Supernova driven
gravito-hydro dynamical model



WARNING: this is not a star
formation simulation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = \dot{n}_{\text{SNe}} M_{\text{ej}}, \quad (1)$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + P \mathbb{I}) = -\rho \nabla \phi + \dot{n}_{\text{SNe}} \mathbf{p}_{\text{SNe}}(Z, n_{\text{H}}), \quad (2)$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot [\rho (e + P) \mathbf{u}] = -n_{\text{H}}^2 \Lambda - \quad (3)$$

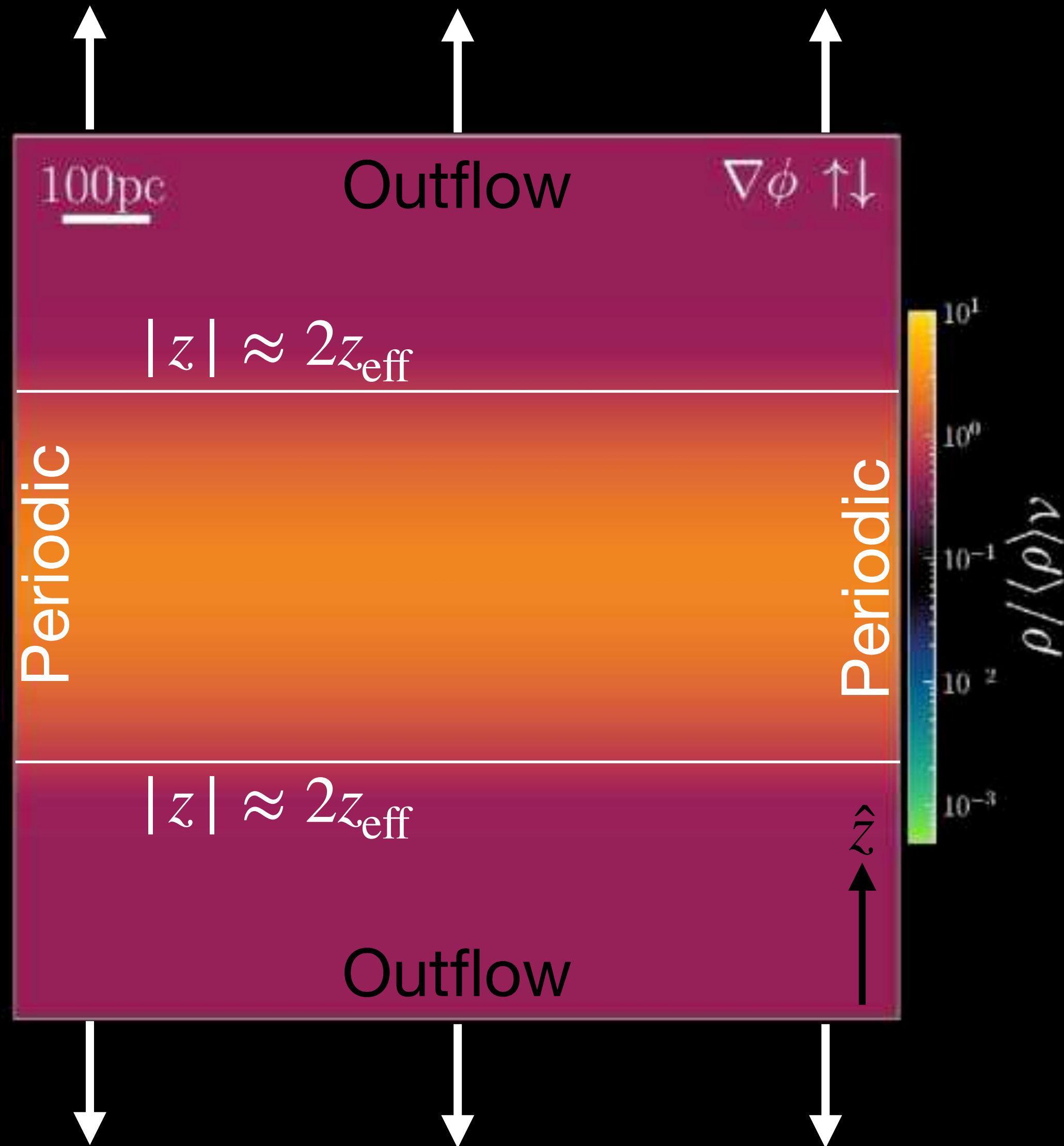
$$\rho \mathbf{u} \cdot \nabla \phi + \dot{n}_{\text{SNe}} \left[E_{\text{th,SNe}}(Z, n_{\text{H}}) + \frac{\mathbf{p}_{\text{SNe}}(Z, n_{\text{H}})^2}{2(M_{\text{ej}} + M_{\text{swept}})} \right], \quad (4)$$

$$e = \epsilon + \frac{u^2}{2}, \quad P = \frac{2}{3} \rho e, \quad (5)$$

$$N_{\text{grid}}^3 = 1024^3 \implies \Delta x \sim 1 \text{ pc}$$

$$\text{numerical diss.} \implies \Delta x \sim 10 \text{ pc}$$

The simplest SNe-driven simulations possible



Static gravitational potential

$$\phi(z) = \underbrace{2\pi G \Sigma_* \left(\sqrt{z^2 - z_0^2} - z_0 \right)}_{\text{stratified disk}} + \underbrace{\frac{2\pi G \rho_{\text{halo}}}{3} z^2}_{\text{spherical halo}}$$

Supernova driving prescription

(following 1D evolution models for momentum and energy deposition in Martizzi + 2016)

$$\dot{n}_{\text{SNe}} = \frac{\dot{\Sigma}_*}{2z_{\text{eff}} 100 M_{\odot}} \quad \text{1 SN per } 100 M_{\odot} \text{ of SF}$$

$$\dot{\Sigma}_* \propto \Sigma^{1.4}$$

KS relation

$$p(z) = \begin{cases} \frac{1}{4z_{\text{eff}}}, & |z| \leq 2z_{\text{eff}} \\ 0, & |z| > 2z_{\text{eff}} \end{cases}$$

$$p(n | \lambda) = \lambda^n e^{-\lambda} / n!$$

$$\lambda(x, y, z) = \dot{n}_{\text{SNe}} \mathcal{V}_{\text{cell}} \Delta t$$